

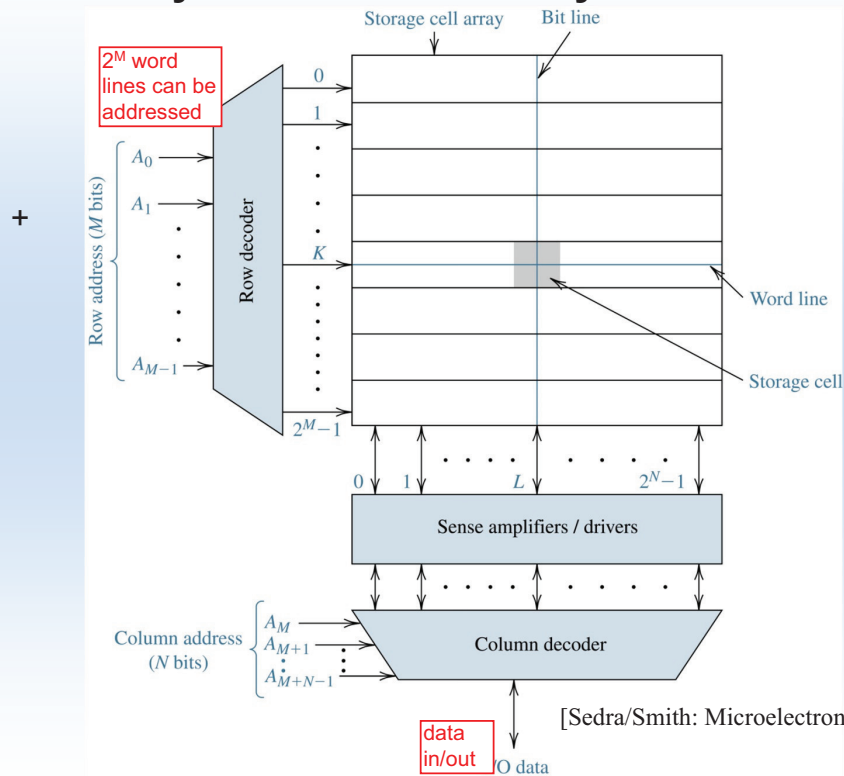
Memory Circuit Design

1. Memory Architectures
2. SRAM/DRAM/FLASH
3. Reliability and Yield

Categories of Semiconductor Memory

Volatile Memory		Nonvolatile Memory
Random Access Memory	Serial Access Memory	EEPROM NAND/NOR FLASH MRAM ReRAM PRAM
DRAM	Shift Registers	
SRAM	FIFO	

Array-structured Memory Architecture



1Gbit Memory device=
 1×10^9 storage cells
 3.2×10^4 rows
 3.2×10^4 columns
 $2^M = 3.2 \times 10^4$
 $M = 15$

propagation delay = $R \times C$

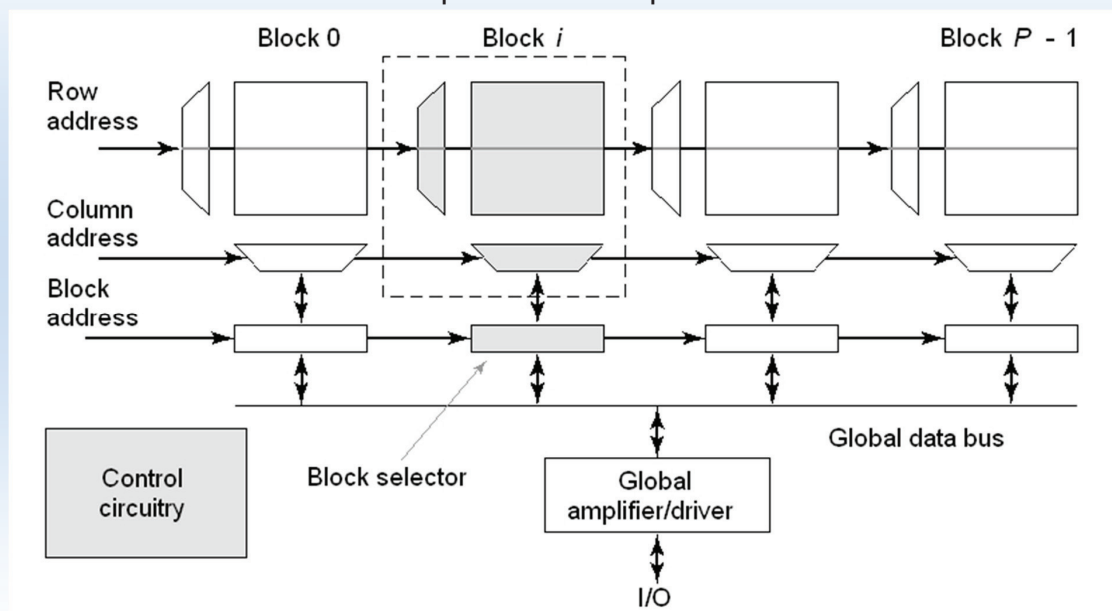
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Hierarchical Memory Architecture

- faster access time
- reduced power consumption



[fig. from: Rabaey et al, Digital Integrated Circuits]

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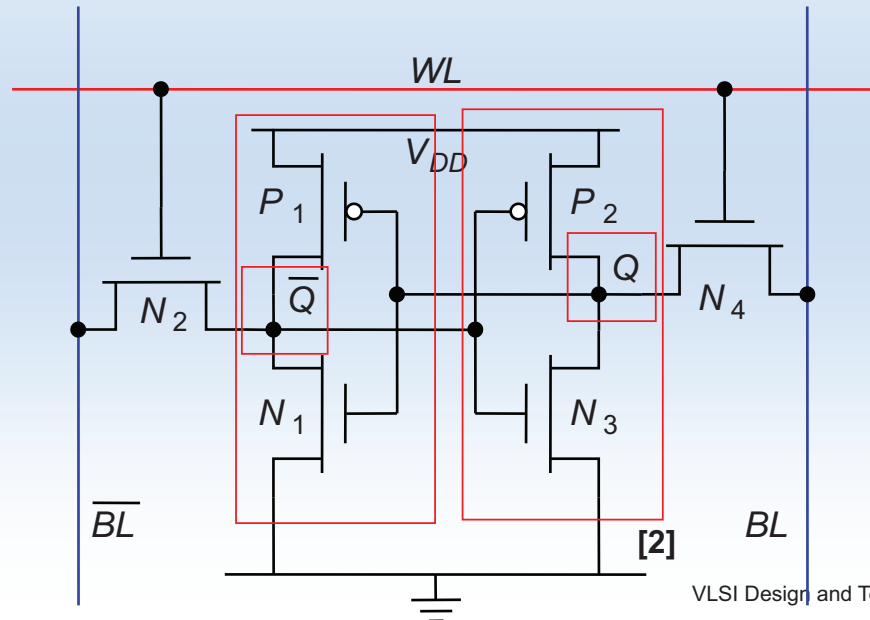
SRAM

Data stored as long as supply is applied

Large (6 transistors/cell)

Fast

6-T SRAM cell



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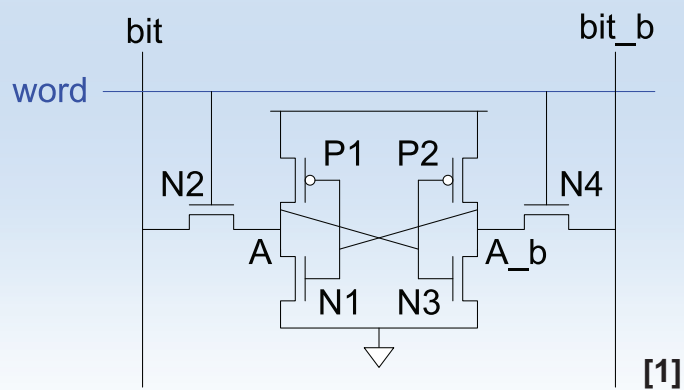
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SRAM

6-T SRAM Cell

Used in most commercial chips

Data stored in cross-coupled inverters



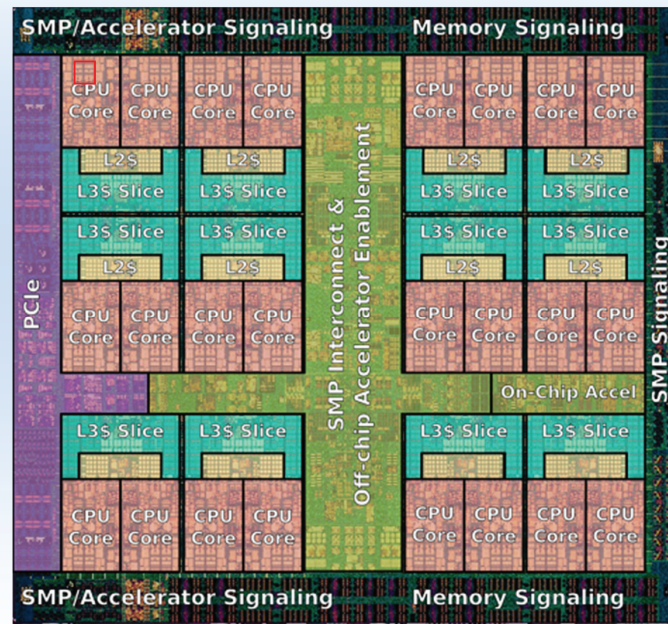
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SRAM

IBM's Power9 processor:
L1+L2 Cache using SRAM, L3 Cache using eDRAM



DRAM
32GBytes=8x32Gbit
chips

SSD

[IBM 2019]

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SRAM

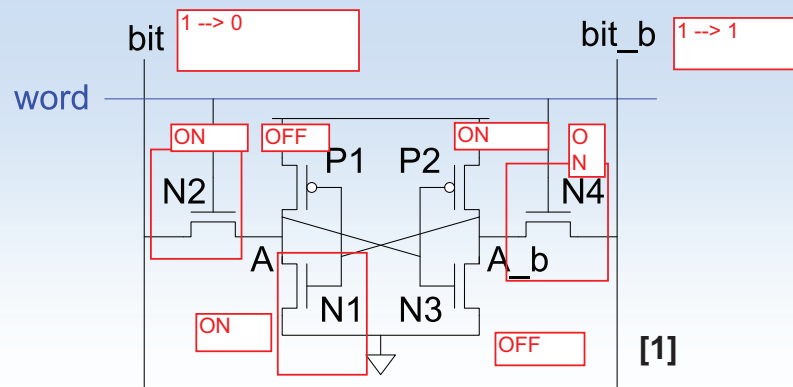
READ operation

- Precharge both bitlines high
- Then turn on wordline
- One of the two bitlines will be pulled down by the cell

Ex: $A = 0, A_b = 1$
bit discharges, bit_b stays high
But A bumps up slightly

- *Read stability*
A must not flip

➔ $N1 \gg N2$



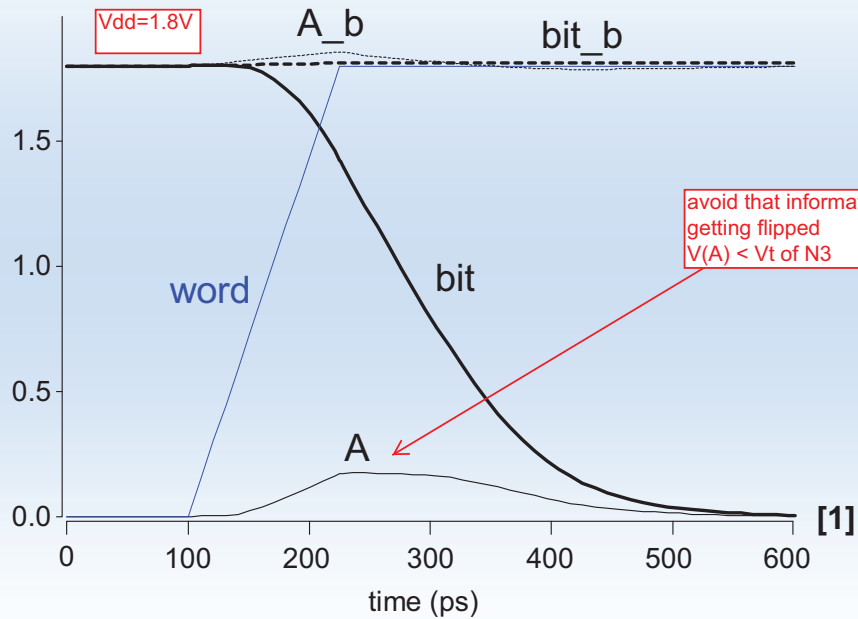
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SRAM

READ operation



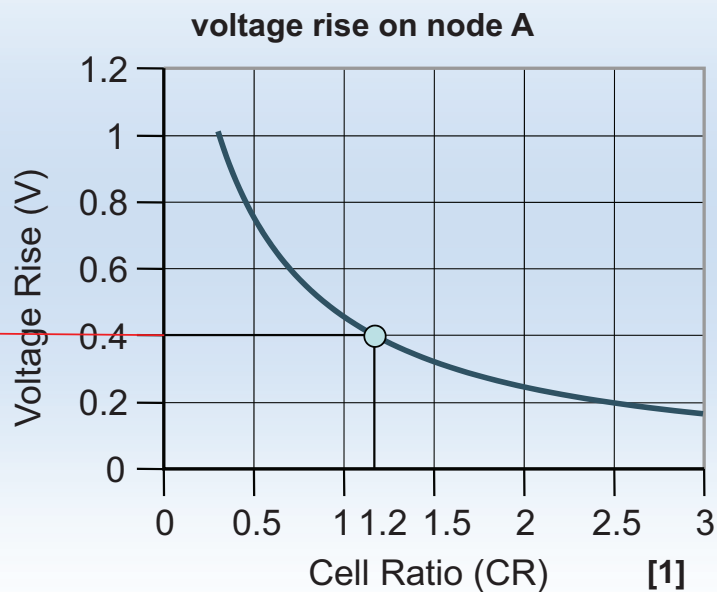
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SRAM

READ operation



$$CR = \frac{W_1/L_1}{W_2/L_2}$$

Voltage on node A must not rise above threshold voltage of transistor N3

→ CR > 1.2 for read stability

Read out 1 results in equal equation for read stability of transistor N3 and N4
 $CR = W_3/W_4 > 1.2$

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SRAM

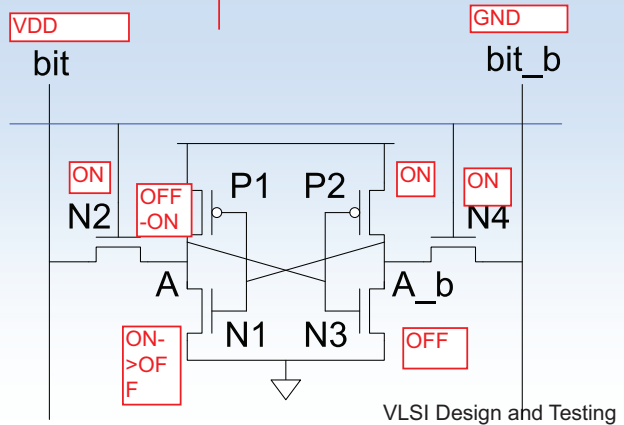
WRITE operation

- Drive one bitline high, the other low
- Then turn on wordline
- Bitlines overpower cell with new value
- Ex: $A = 0$, $A_b = 1$, $\text{bit} = 1$, $\text{bit}_b = 0$
- bitline is unable to force A high through N2, so force A_b low through N4, then A rises high

• writeability

Must overpower feedback inverter

➔ $N4 \gg P2$



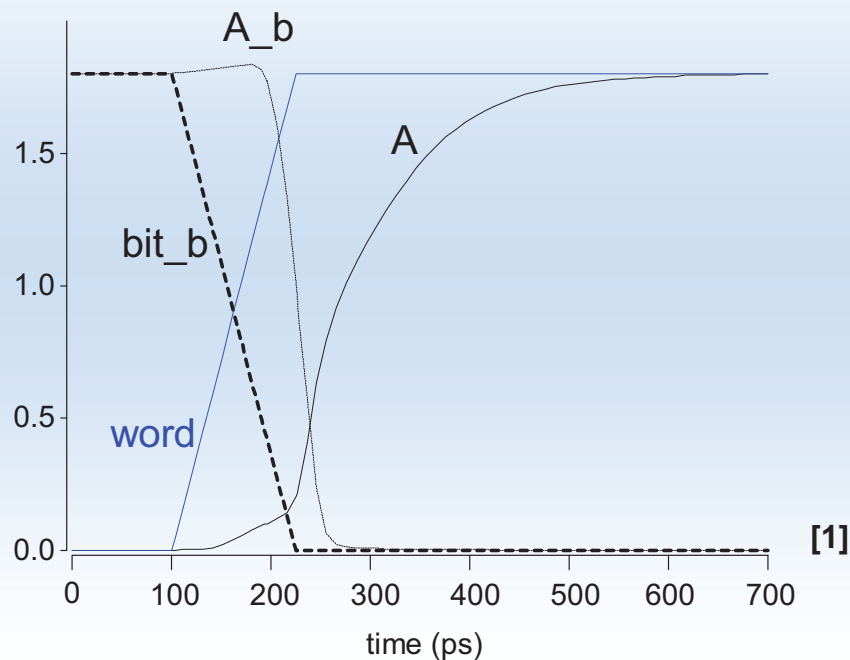
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SRAM

WRITE operation



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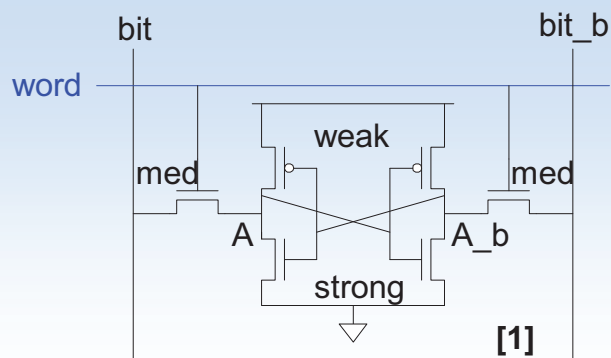
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SRAM

Transistor Sizing

- Cell information must not flip during read: High bitlines must not overpower inverters during reads. So strong NMOS transistors in the cross coupled inverters are needed!
- But low bitlines must write new value into cell: So PMOS transistors must be weak!



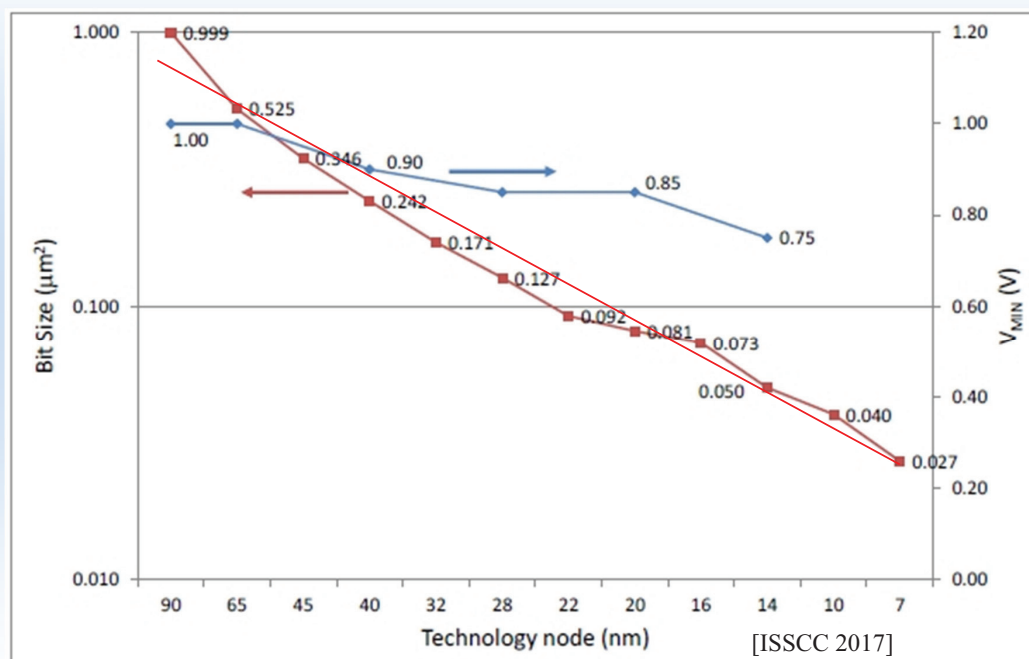
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SRAM

Scaling trend: Cell size and V_{min}

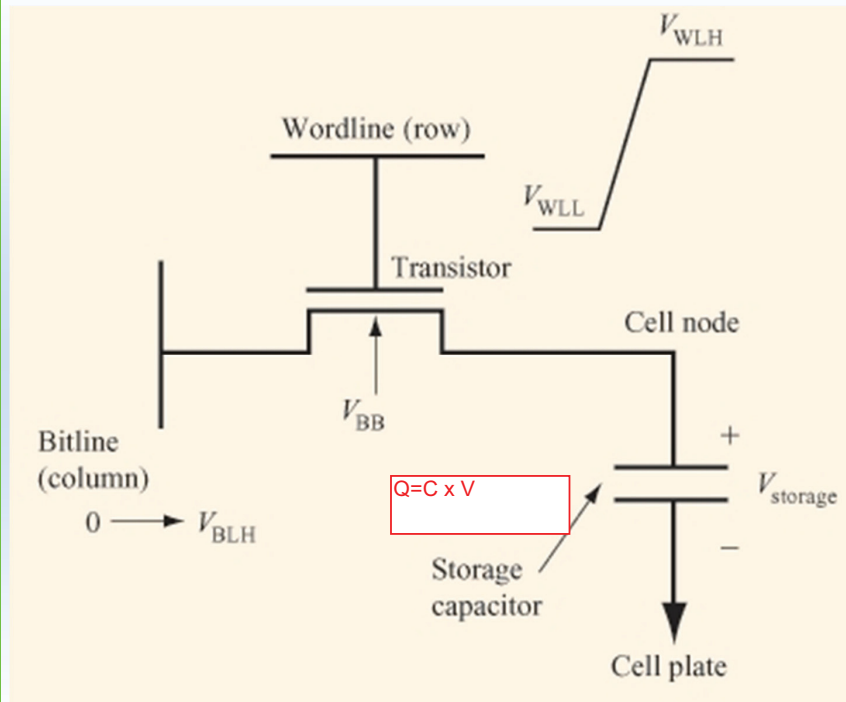


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1-T DRAM cell



Basic functionality:

Binary coded information is stored as charge on a capacitor

Because of leakage current of the transistor, charge must be periodically refreshed!

Dynamic Random Access Memory

[Mandelman01]

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Trench vs. Stack DRAM capacitor

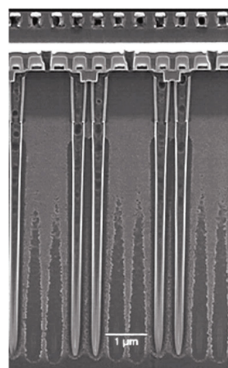
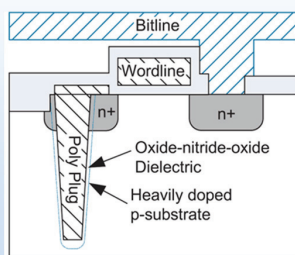
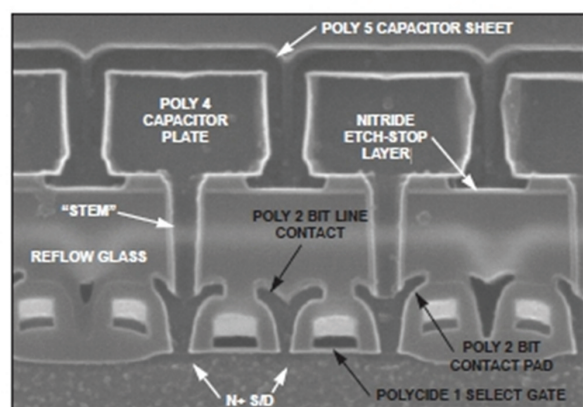


FIG 11.27 Trench capacitor

Capacity proportional to the plate area

[Weste05]



20844

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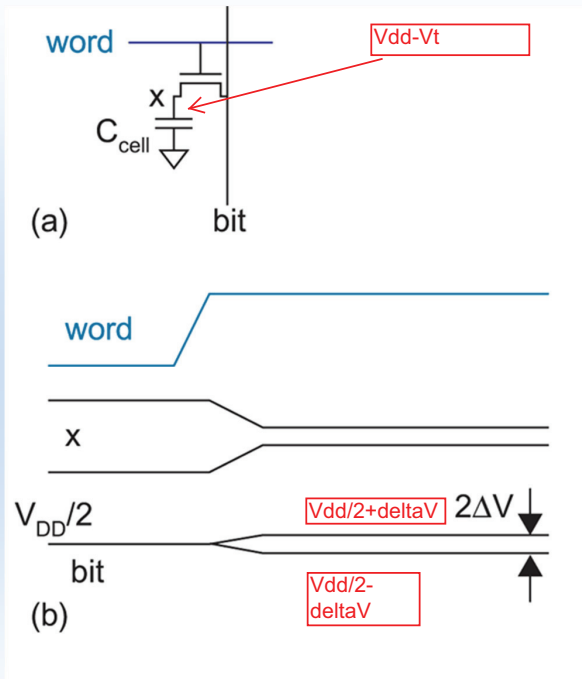


FIG 11.26 DRAM cell read operation

[Weste05]

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DRAM read

1. bitline is precharged to $V_{DD}/2$
2. activate wordline (rise wordline to high) $\Rightarrow$ all cells connected to this wordline are accessed
3. capacitor shares its charge with the bitline causing a voltage change ΔV that can be sensed

$$\Delta V = \frac{V_{DD}}{2} \cdot \frac{C_{Cell}}{C_{Cell} + C_{bitline}}$$

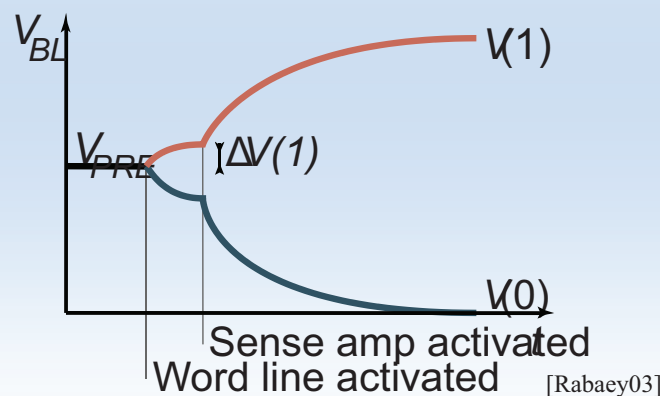
Assuming the voltage on node x is either V_{DD} for "1" or 0 for a "0"

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Conclusions

1. DRAM requires a sense amplifier for each bitline to be functional
2. The read out of the DRAM cell is destructive: after a read operation the original value must be restored! So a refresh is followed after each read operation. Typically during the read out the output of the sense amplifier is written back to the bitline.



Bitline voltage waveform during read operation

[Rabaey03]

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Conclusions

3. The ratio $\frac{C_{Cell}}{C_{Cell} + C_{bitline}}$ is called charge-transfer ratio. For reliability and fast read out it needs to be kept large! So a minimum value of the cell capacitance of about 30fF is needed. To fit this capacitance in as small an area as possible is one of the key challenges in DRAM design!! (see figure trench cell)
4. When writing a “1” into the cell a threshold voltage is lost, which reduces the available charge. This charge loss can be avoided by “bootstrapping” the wordline voltage higher than Vdd!

Bitline voltage swing during read operation

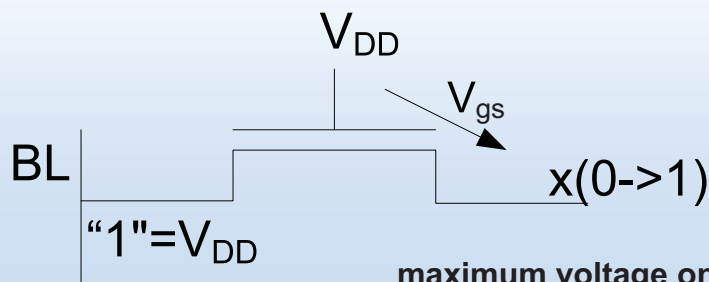
$$\Delta V = \frac{V_{DD}}{2} \cdot \frac{C_{Cell}}{C_{Cell} + C_{bitline}}$$

with: $V_{DD} = 2.5V$
 $C_{cell} = 50 fF$
 $C_{bitline} = 1 pF$

$$\Delta V = 60 mV$$

small voltage swing on the bitline!

DRAM Write Operation

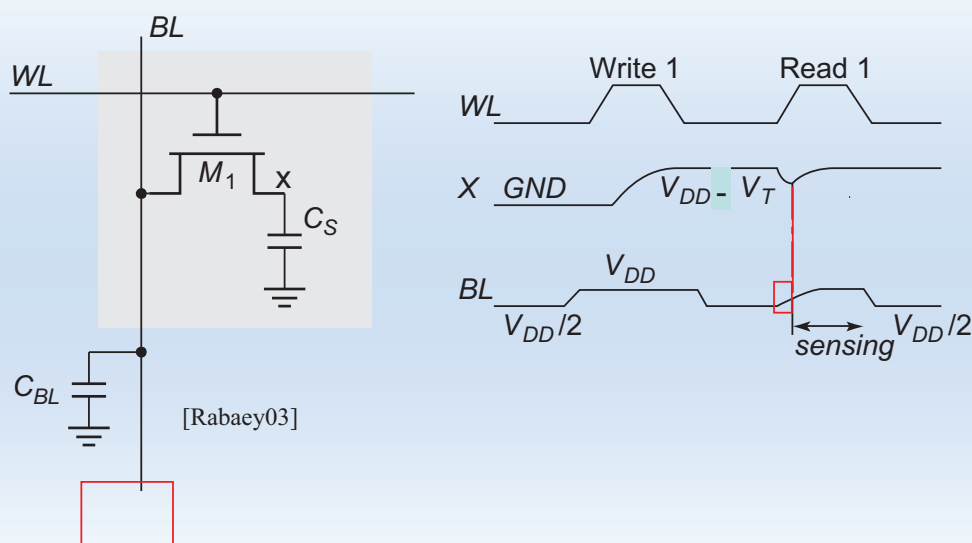


maximum voltage on node x is:

$$V_x = V_{cell} = V_{DD} - V_t$$

When writing a "1" into the cell a threshold voltage is lost, which reduces the available charge. This charge loss can be avoided by "bootstrapping" the wordline voltage higher than V_{DD} !

DRAM write-read



Array Architecture

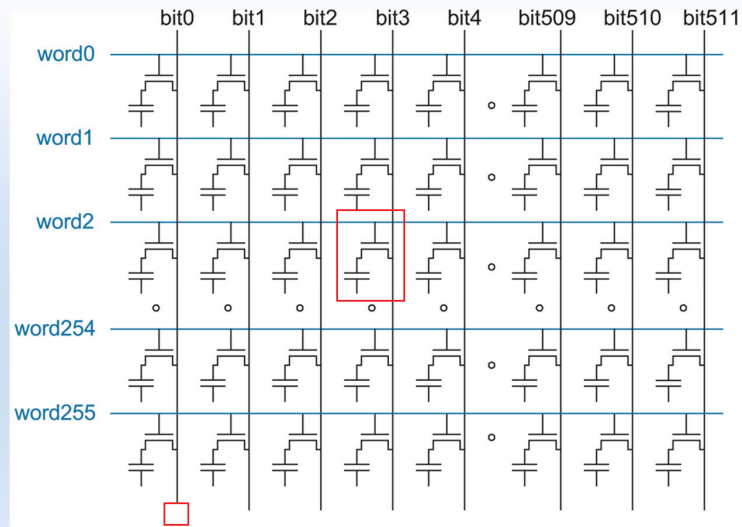


FIG 11.28 DRAM subarray

[Weste05]

Subarray size is a tradeoff between density and performance

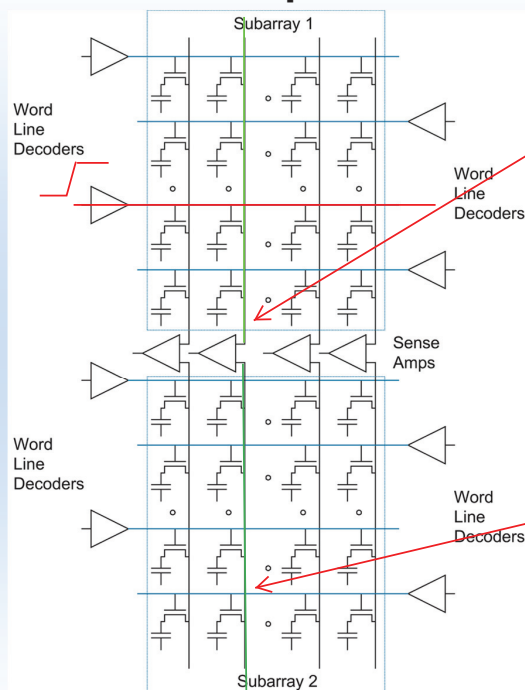
Typical size: 256 words by 512 bits

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Open Bitline Architecture



$$\begin{aligned} &V_{dd}/2 + V_{noise} + \Delta V \\ &V_{dd}/2 + V_{noise} - \Delta V \end{aligned}$$

In the open bitline architecture the sense amplifier receives one bitline from each of the two subarrays

Problem: noise on bitline appears as differential noise at the sense amplifier!

$$V_{dd}/2$$

FIG 11.29 Open bitlines

[Weste05]

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Folded Bitline Architecture

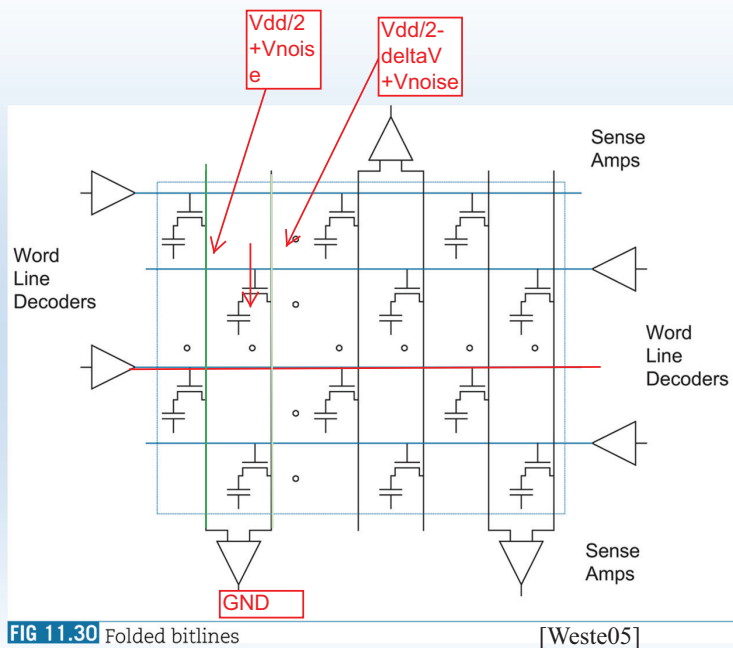


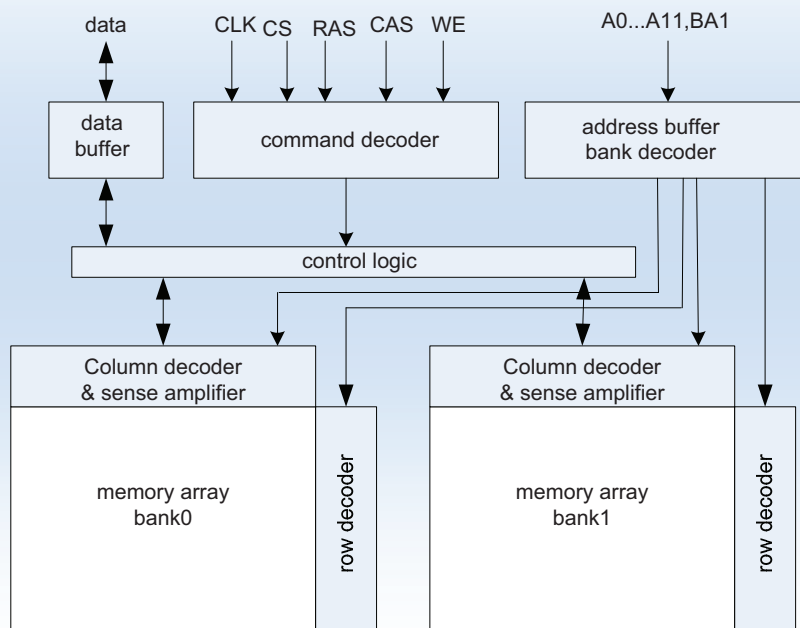
FIG 11.30 Folded bitlines

[Weste05]

Adjacent bitlines are organized in pairs as inputs to the sense amplifier: WL-to-BL noise couples equally onto the two adjacent bitlines. So noise appears as common mode noise and is rejected by the sense amplifier!

SDRAM

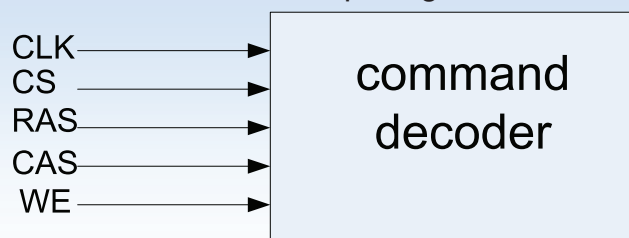
SDRAM commands, addresses and data are latched at the rising edge of a clock signal CLK!



SDRAM

- Basic SDRAM operation:
- activate (ACT)
 - read (RD) followed by a precharge (PRE)
 - write (WR) followed by precharge (PRE)
 - refresh (REFR) of all cells for one row address
 - no operation (NOP) used as time fillers
 - mode register set

SDRAM command input signals:



SDRAM operations

CLK	CS	RAS	CAS	WE	operation
rising edge	L	L	H	H	activate
rising edge	L	H	L	H	read
rising edge	L	H	L	L	write
rising edge	L	L	H	L	precharge
rising edge	L	H	H	L	burst stop
rising edge	L	L	L	H	refresh
rising edge	L	L	L	L	mode register set
rising edge	L	H	H	H	no operation

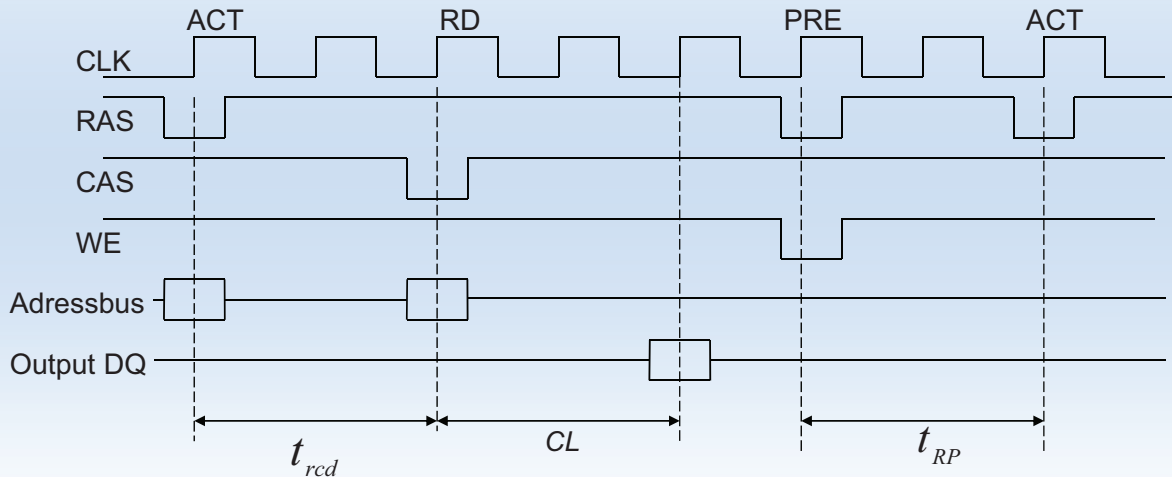
CS = chip select

RAS = row address select

CAS = column address select

WE = write enable

SDRAM read operation



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SDRAM read operation

ACTIVATE Command

The ACTIVATE command is used to open (or activate) a row in a particular bank for a subsequent access.

The value on the BA0-BA2 inputs selects the bank, and the address provided on inputs A0-A15 selects the row. This row remains active (or open) for accesses until a precharge command is issued to that bank. A PRECHARGE command must be issued before opening a different row in the same bank.

PRECHARGE Command

The PRECHARGE command is used to deactivate the open row in a particular bank. The bank(s) will be available for a subsequent row activation a specified time (t_{RP}) after the PRECHARGE command is issued. Once a bank has been precharged, it is in the idle state and must be activated prior to any READ or WRITE commands.

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SDRAM read operation

For every row address a wordline WL is selected and all cells along a wordline are read out to sense amplifiers and **refreshed** with a write back during sensing.

CAS-latency (CL) defines the number of clock cycles the data is available at the outputs after the column address is applied. The CL can be set by the user applying the mode register set command.

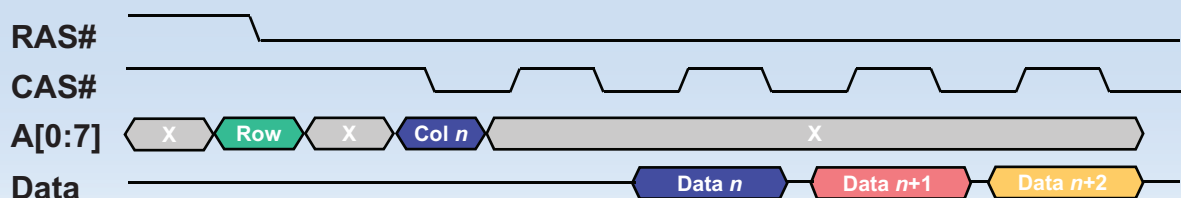
RAS-to-CAS delay (trcd) is the minimum time (number of clock cycles) between the activate command and the first read or write command.

trp is the time the cell is disconnected from the sense amplifier and the sense amplifiers are equalized to be ready for the next access.

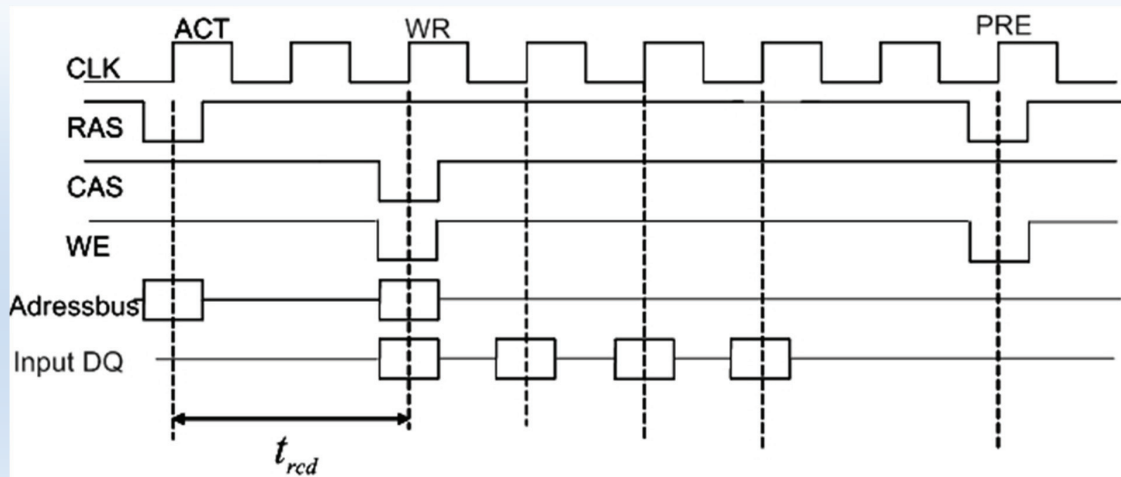
SDRAM Burst operation

• Burst oriented read/write access

- Generates consecutive column address by itself
- Burst length is programmable: 1, 2, 4, 8 and full-page
- May end full-page by Burst Stop command



SDRAM write operation

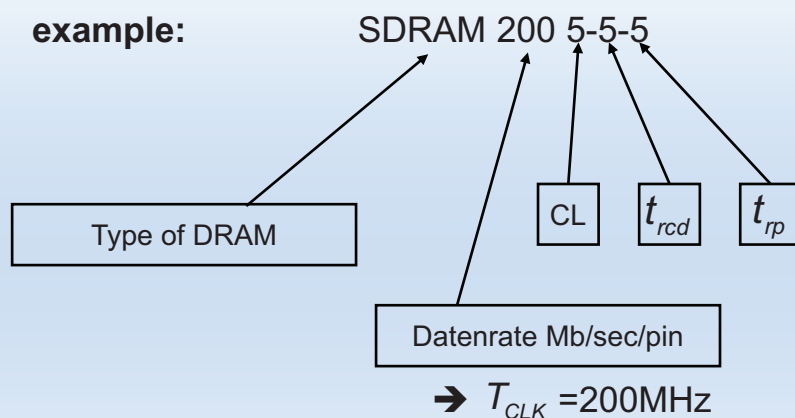


Burst Length = 4

Burst Length can be programmed by Mode Register Set command

SDRAM specification

example:



SDRAM specification

DDR (double data rate) SDRAM:

DDR SDRAM latches data at the rising and the falling clock edge!

double the data transfer rate

HYB18TC1G[80/16]0AF
1-Gbit DDR2 SDRAM

TABLE 1

Performance table for -3S

Product Type Speed Code			-3S	Unit
Speed Grade			DDR2-667D 5-5-5	—
Max. Clock Frequency	@CL5	f_{CK5}	333	MHz
	@CL4	f_{CK4}	266	MHz
	@CL3	f_{CK3}	200	MHz
Min. RAS-CAS-Delay		t_{RCD}	15	ns
Min. Row Precharge Time		t_{RP}	15	ns

data rate=667MHz/pin
clk=333MHz
T=3ns

DDR 667 5-5-5

- $T_{CLK} = 333\text{MHz}$
- $CL = 15\text{ns}$
- $t_{RCD} = 15\text{ns}$
- $t_{RP} = 15\text{ns}$

[Infineon Technologies]

DDR3 Technology Overview

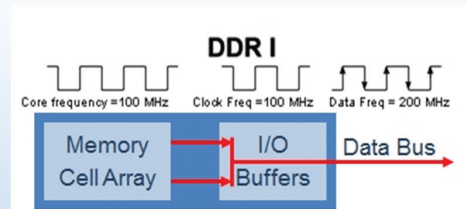
Feature	DDR3	DDR2	Benefit	Impact
Signaling	Enhancements	-	Performance	Enable higher speed signaling on the system
Pre-fetch	8n	4n	Performance	Doubles DRAM I/O bandwidth at same core frequency
Voltage	1.5 V	1.8 V	Power	Lower active, idle, standby power
Power Saving Modes	DLL-off, ASR*	-	Power	Lower idle and self refresh (S3) power

*Optional Mode
ASR = Automatic Self Refresh (Temperature Compensated)

DDR SDRAM Prefetch

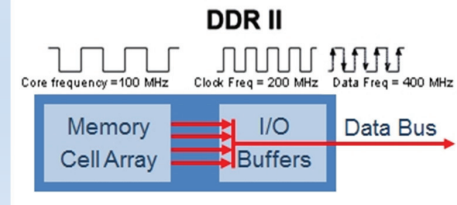
2n-prefetch of DDR I:

2 datawords are stored after sensing, I/O data bus operates at 2x data frequency



4n-prefetch of DDR II:

4 datawords are stored after sensing, cell array works at same frequency, I/O data bus operates at double clock frequency, resulting in 4x data frequency



prefetch buffer size of DDR SDRAM:

DDR SDRAM's prefetch buffer size is 2n (two datawords per memory access)

DDR2 SDRAM's prefetch buffer size is 4n

DDR3,DDR4 SDRAM's prefetch buffer size is 8n

DDR5 SDRAM's prefetch buffer size is 16n

DDR SDRAM Specifications

	DDR	DDR2	DDR3	DDR4	DDR5*
Prefetch-Buffer	2	4	8	8	16
Supply-Voltage [V]	2,5	1,8	1,5	1,2	1,1
I/O-Datarate [Mb/s]	200-400	400-1066	800-2133	2133-4266	3200-8400
I/O-Bus Frequency [MHz]	100-200	200-533	400-1066	1066-2133	1600-4200
Cell-Array Frequency [MHz]	100-200	100-267	100-267	267-533	200-525

*released
in 07/2020

DDR4 SDRAM

DDR4-2133-15-15-15

I/O Datarate: 2133 Mbit/s per pin

I/O-Bus Frequency: 1066,5 MHz

Cell-Array Frequency: 266,6 MHz (8n-prefetch)

CL=15, RAS-CAS-delay=15, Row-Precharge-Time=15

$$CL = 15 \cdot 1 / 1066,5 \text{ MHz} = 14 \text{ ns} = t_{rcd} = t_{rp}$$

data transfer rate of a module [MByte/s]=

$$\text{cell-array frequency [MHz]} \times 64\text{-bit I/O} \times 8\text{-bit Prefetch} / 8 \text{ bit} =$$

$$= 17064 \text{ MB/s} = 17 \text{ GB/s}$$

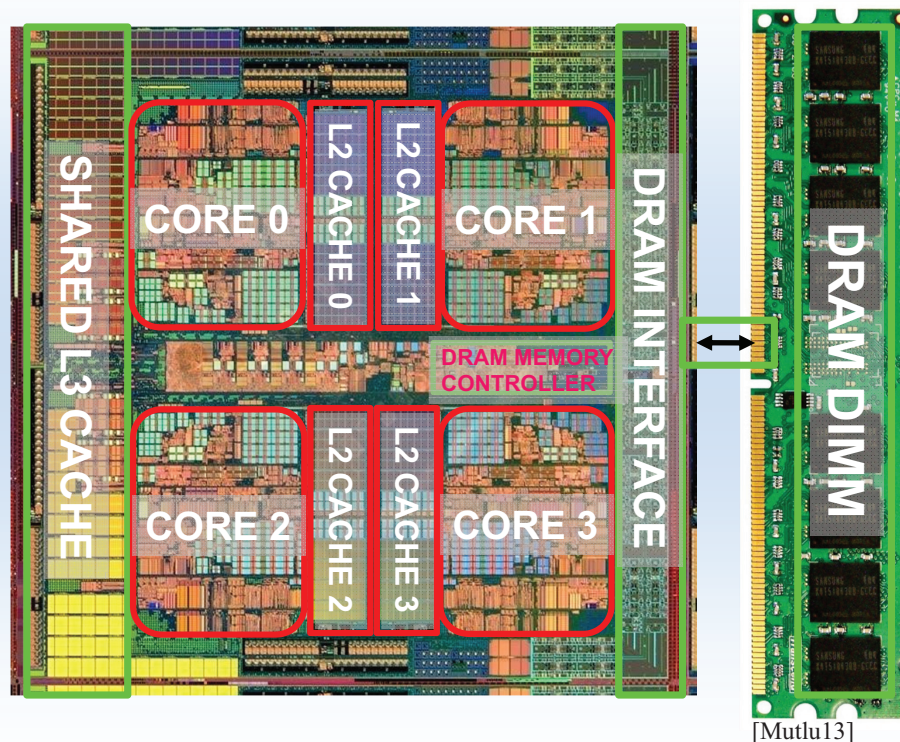
DDR4-1600	PC4-12800	200 MHz	800 MHz	1600 MHz	1,2 V	12,8 GB/s
DDR4-2133	PC4-17000	266 MHz	1066 MHz	2133 MHz	1,2 V	17,0 GB/s
DDR4-2666	PC4-21300	333 MHz	1333 MHz	2666 MHz	1,2 V	21,3 GB/s
DDR4-3200	PC4-25600	400 MHz	1600 MHz	3200 MHz	1,2 V	25,6 GB/s

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SRAM+DRAM in the System



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DRAM Memory Controller

Ensure correct operation of DRAM (refresh and timing)

Service DRAM requests

Translate requests to DRAM command sequences

Buffer and schedule requests to improve performance

Reordering, row-buffer, bank and rank scheduling

Manage power consumption and thermals in DRAM

Turn on/off DRAM chips, manage power modes

DRAM Memory Controller on CPU chip

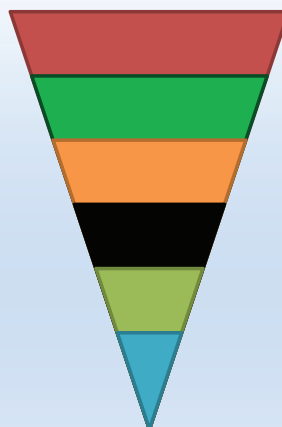
+ Reduced latency for main memory access

+ Higher bandwidth between cores and controller

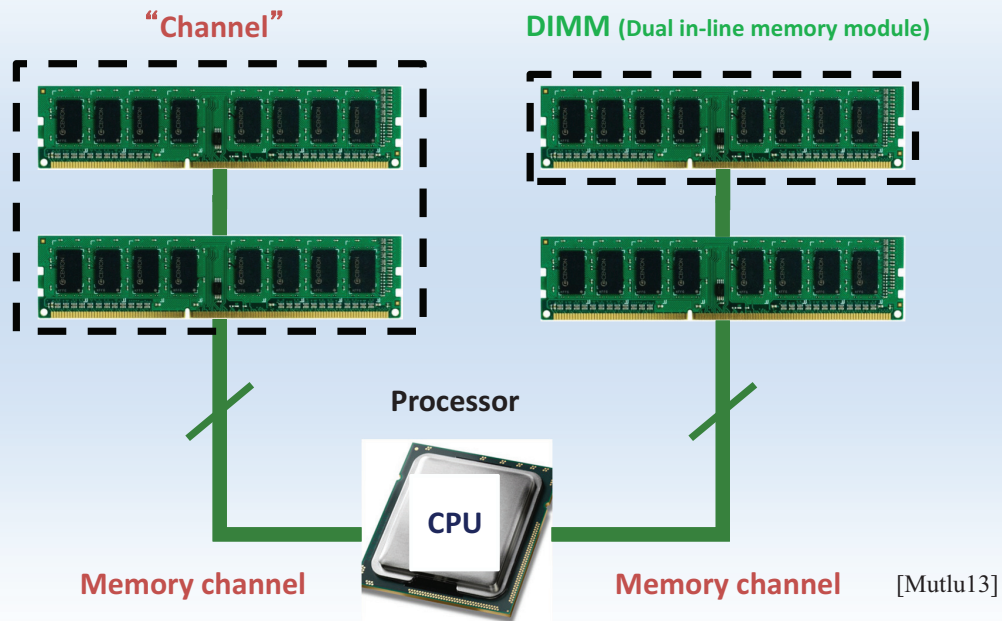
[Mutlu13]

DRAM Subsystem Organization

- Channel
- DIMM
- Rank
- Chip
- Bank
- Row/Column



DRAM Subsystem



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DIMM

- DIMM: Dual In-line Memory Module
 - A small circuit board that holds memory chips



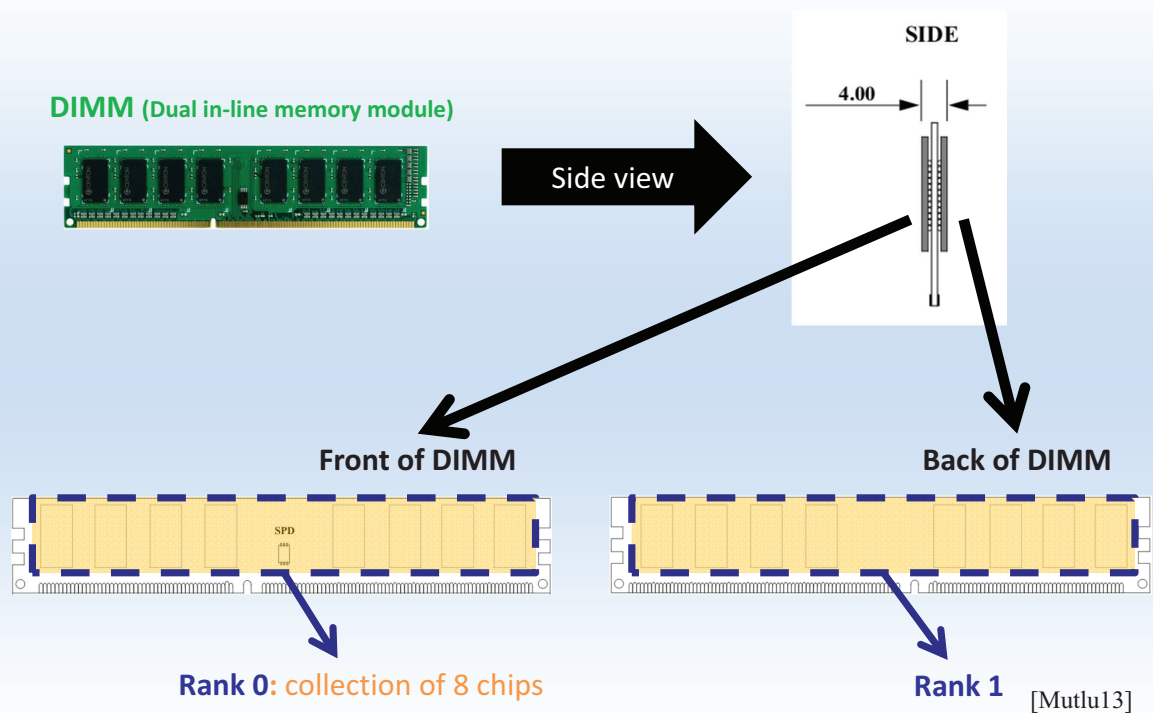
- 64-bit wide data path (72 bit with parity)
 - Single sided: 9 chips, each with 8 bit data bus (9th chip for parity check)
 - Dual sided: 18 chips, each with 4 bit data bus
 - Data BW: 64 bits on each rising and falling edge of the clock
- Other pins
 - Address – 14, CS, RAS, CAS, WE – 4, VDC – 17, Gnd – 18, clock – 4, ...

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115

DIMM

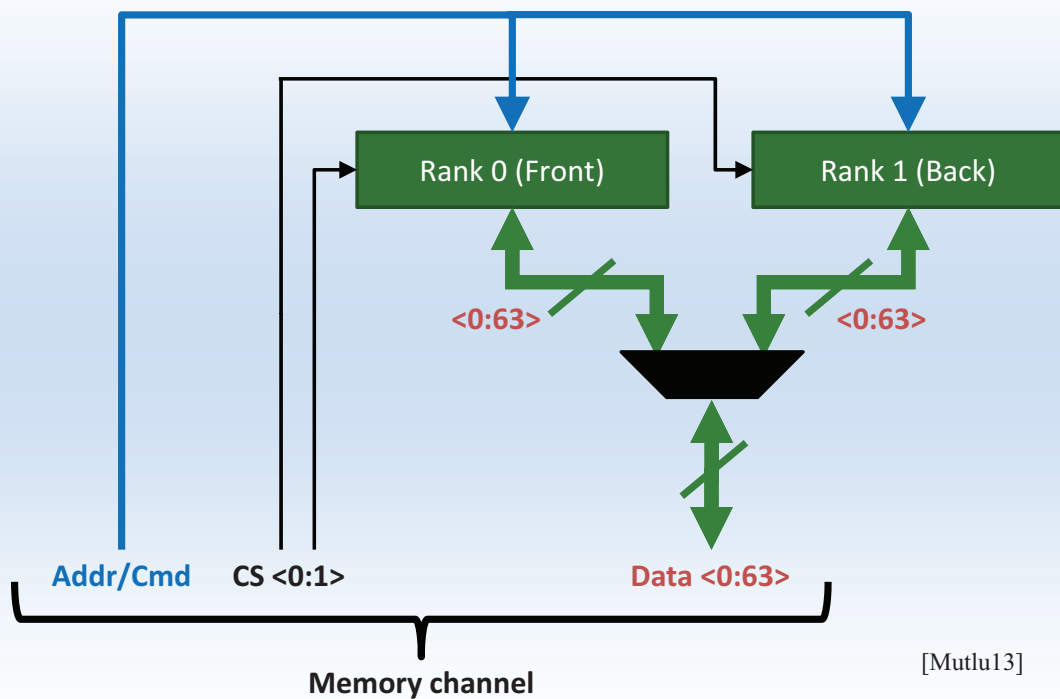


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Access of a DIMM

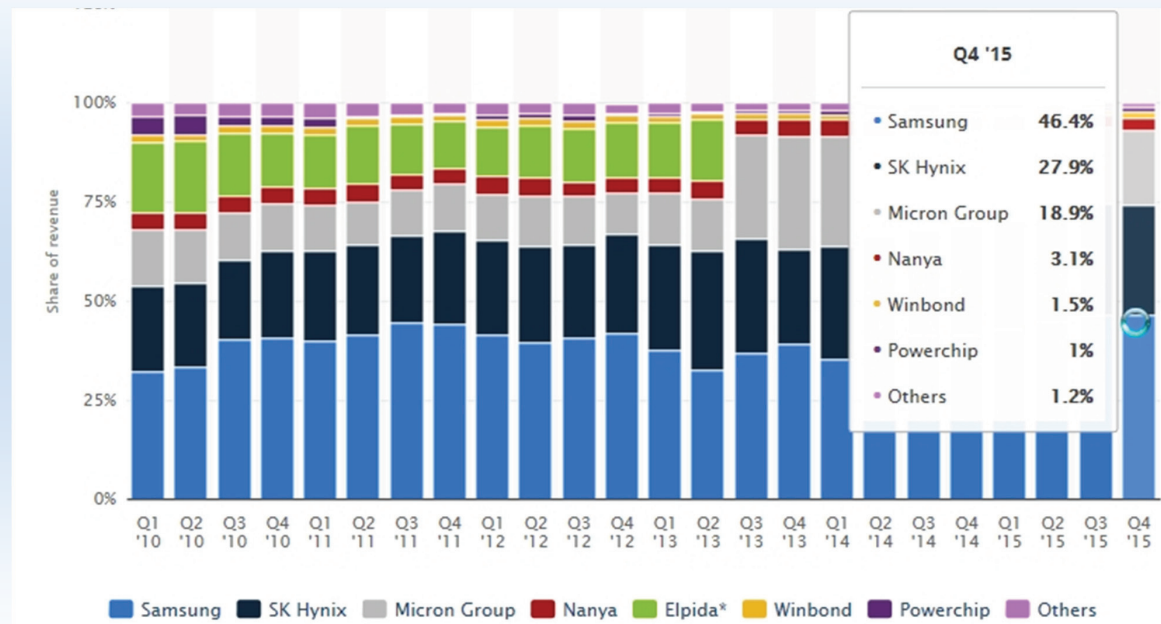


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Market Share by DRAM chip vendors



[Statista 2016]

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Trend: eDRAM

Embedded DRAM (eDRAM) is mainly used as L3+L4 cache for graphics engine on CPU (Intel). IBM z15 microprocessor uses eDRAM in all levels of cache (L1-L3). Advantage over SRAM: - high density (large cache memory capacity possible)

Use of eDRAM in various products	
Product name	Amount of eDRAM
IBM z15	256+ MB
IBM's System Controller (SC) SCM, with L4 cache for the z15	960 MB
Intel Haswell , Iris Pro Graphics 5200 (GT3e)	128 MB
Intel Broadwell , Iris Pro Graphics 6200 (GT3e)	128 MB
Intel Skylake , Iris Graphics 540 and 550 (GT3e)	64 MB
Intel Skylake , Iris Pro Graphics 580 (GT4e)	64 or 128 MB
Intel Coffee Lake , Iris Plus Graphics 655 (GT3e)	128 MB
PlayStation 2	4 MB
PlayStation Portable	4 MB
Xbox 360	10 MB
Wii U	32 MB

[Wikipedia 2020]

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119

References

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by J. A. [Mandelman](#), R. H. [Dennard](#), G. B. [Bronner](#), J. K. [DeBrosse](#), R. [Divakaruni](#), Y. [Li](#), and C. J. [Radens](#), IBM research journal, 2001.

[Weste05] N.H.E. Weste, D. Harris "CMOS VLSI Design", 3rd edition, Addison Wesley, 2005.

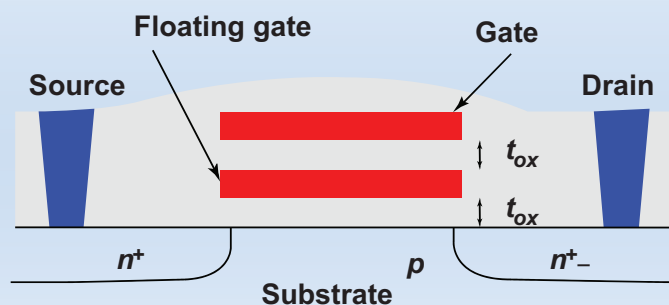
[Rabaey03] J.M. Rabaey, A. Chandrakasan, B. Nikolic "Digital Integrated Circuits", 2nd edition, Prentice Hall, 2003.

[Mutlu13] O. Mutlu, "DRAM basics and DRAM scaling", lecture at Carnegie Mellon University, 2013.

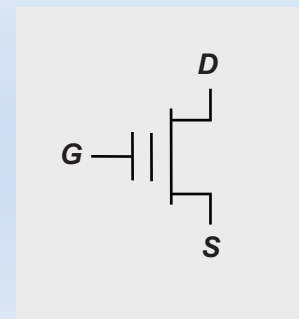
[ISSCC17] Intern. Solid-State Circuit Conference "ISSCC 2017 Trends", San Francisco, 2017.

Non-volatile Memory: FLASH

Floating-Gate Transistor (FAMOS)



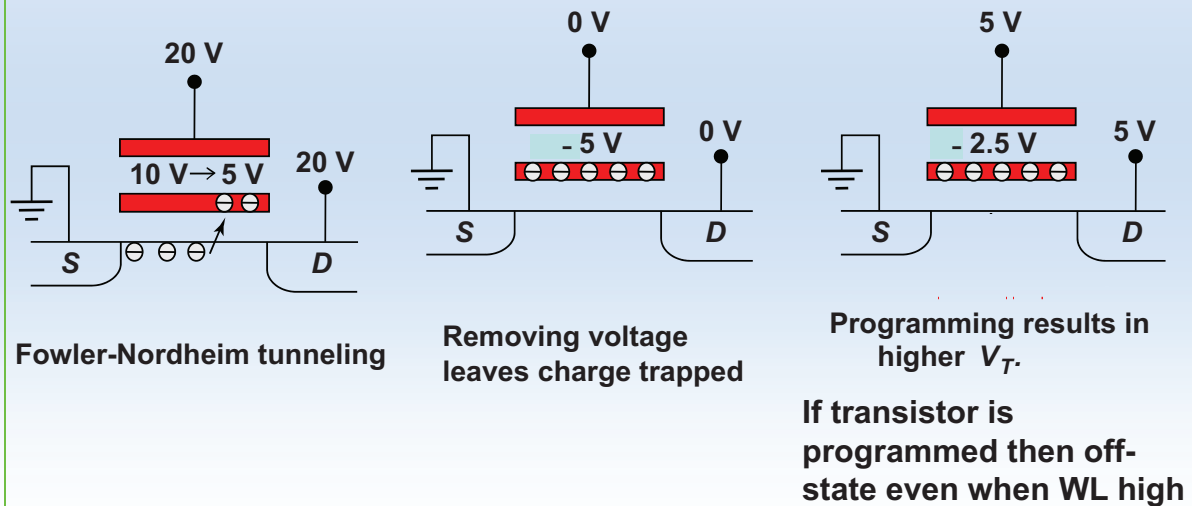
Device cross-section [2]



Schematic symbol

Non-volatile Memory: FLASH

Floating-Gate Transistor Programming



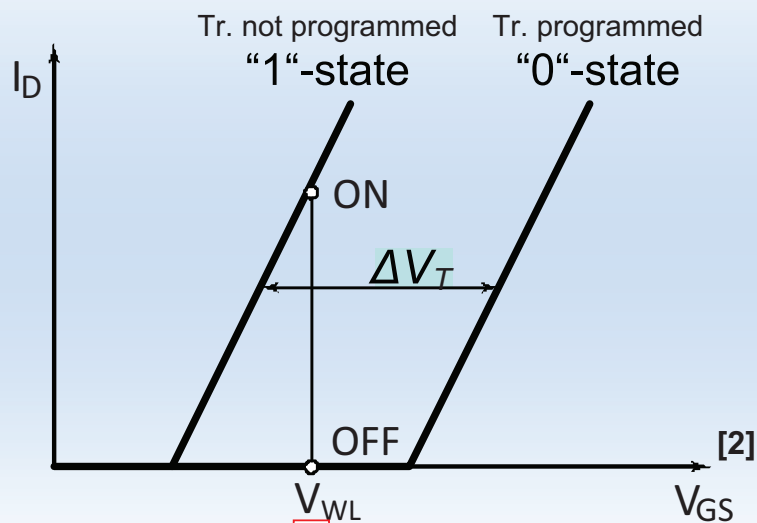
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Non-volatile Memory: FLASH

Charge injected onto the floating gate increases the threshold voltage



Applying a wordline voltage V_{WL} results in either current to flow ("1" state) or not ("0" state)

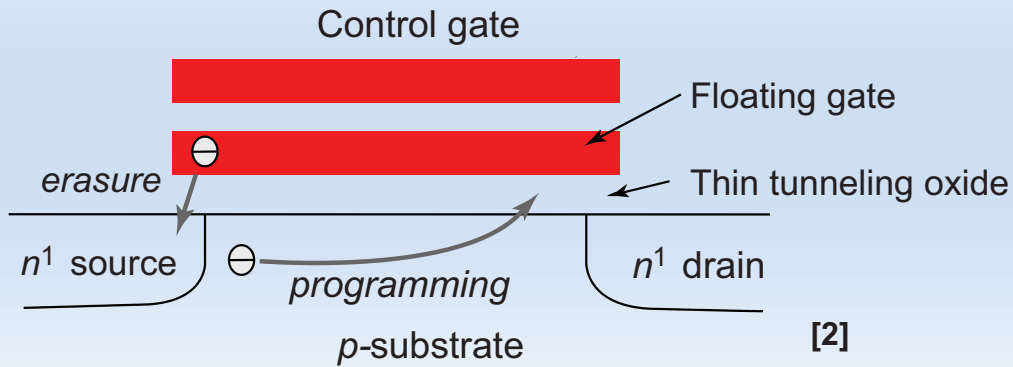
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FLASH Memory

All nonvolatile memory is based on the floating gate approach. Different classes can be identified based on the erasure mechanism. For the most popular FLASH memory erasure is performed using *Fowler-Nordheim tunneling*.

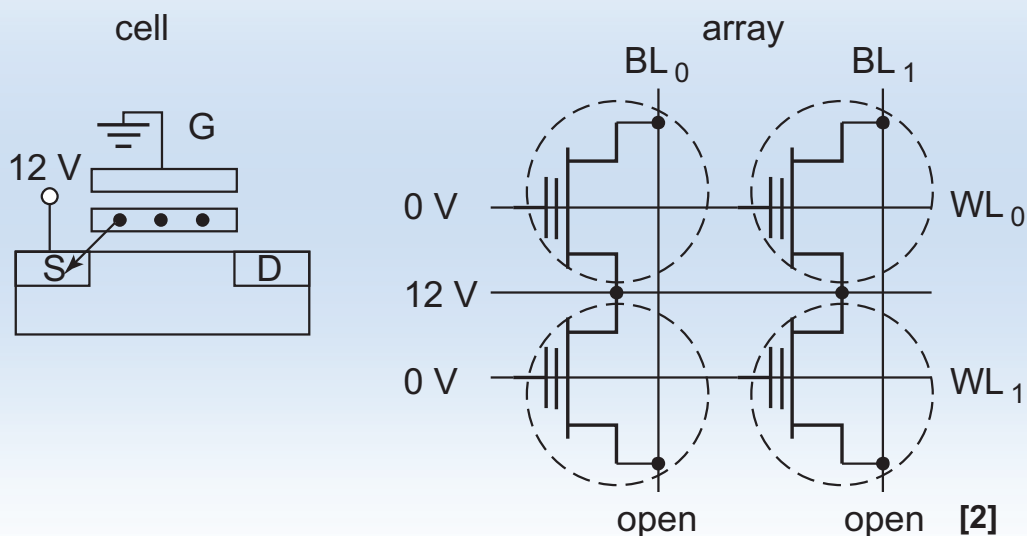


Erasure is performed for the complete chip or a subsection: This makes monitoring of the threshold voltage during erasure possible and the erasure time can be adjusted dynamically by a control hardware on chip.

NOR FLASH Memory

Erasure Operation

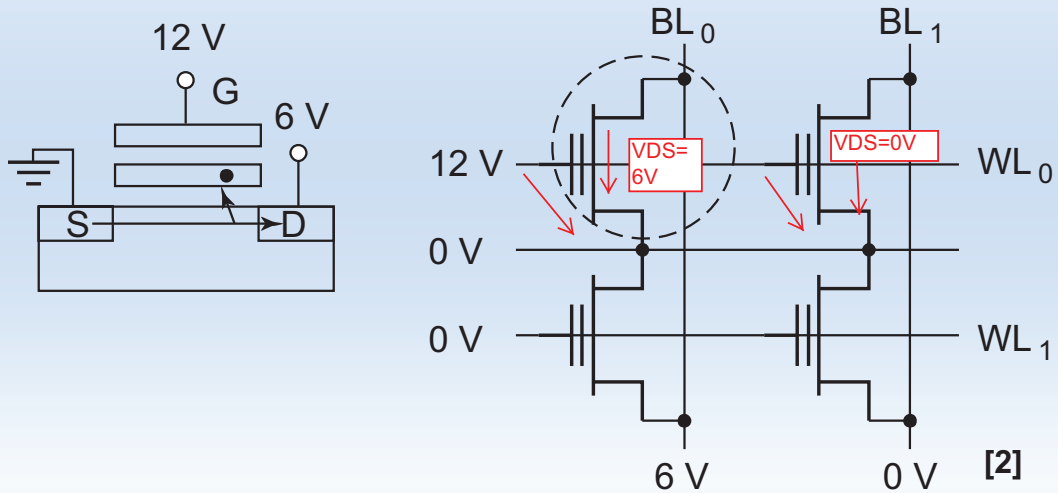
for complete subarray



NOR FLASH Memory

Write Operation

on a single cell



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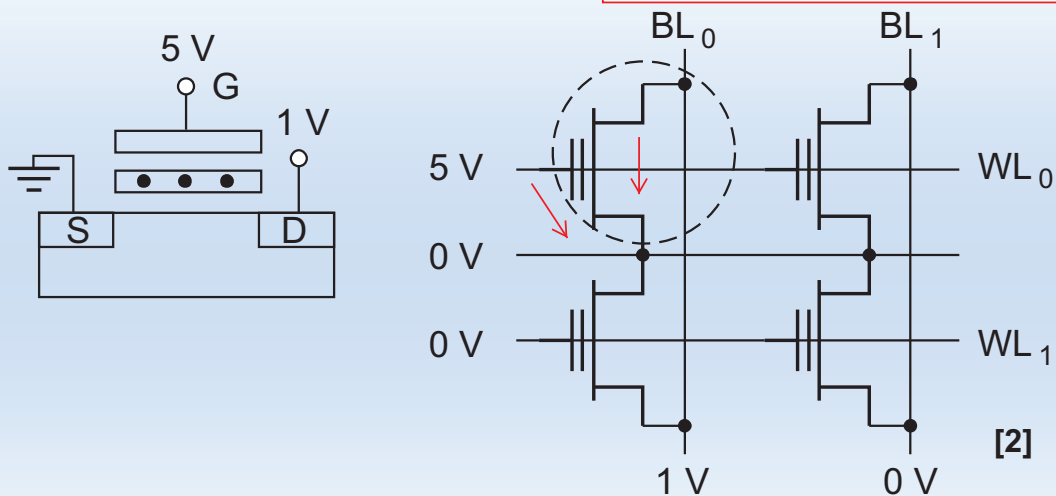
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NOR FLASH Memory

Read Operation

if transistor programmed --> transistor in OFF state --> $I_d=0$ --> $BL=1V$
 if transistor not programmed --> transistor ON --> $I_d>0$ --> $BL=0V$



NOR architecture leads to fast read operation. But erasure and write is slow!

NOR architecture is used when you want to perform mainly read operations: program

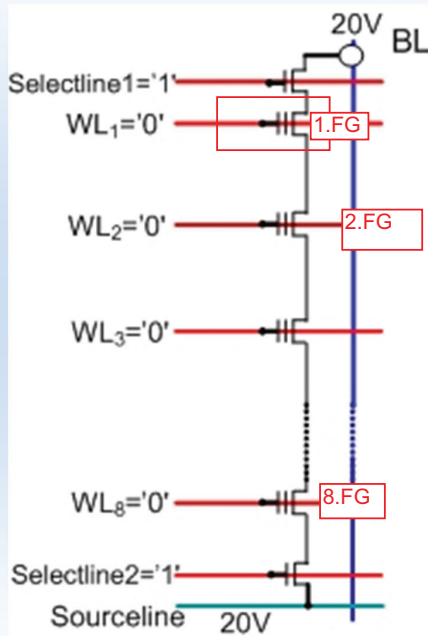
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NAND FLASH Memory

Erasure Operation



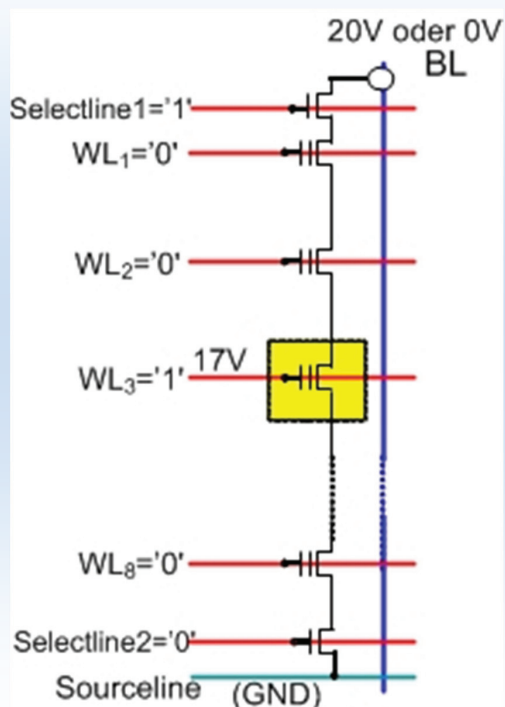
NAND Flash module consists of 8 to 16 floating-gate transistors connected in series. They are connected to bitline BL and Sourceline via two MOS-transistors.

During erasure all floating-gate transistors get nominal, negative threshold voltage: $V_t \approx -3V$

1.step: 1.FG is getting erased -->
threshold voltage set to nominal voltage $V_t = -3V$ --> FG is ON-state
8.FG is getting erased -->
threshold voltage set to nominal voltage $V_t = -3V$ --> FG is ON-state
2.step: 2.FG is getting erased -->
threshold voltage set to nominal voltage $V_t = -3V$ --> FG is ON-state
7.FG is getting erased -->
threshold voltage set to nominal voltage $V_t = -3V$ --> FG is ON-state
3.step: 3.+6. FG is getting erased, and so on.

NAND FLASH Memory

Write Operation



Before writing information into cell, erasure of complete block necessary!

Meaning all floating-gate transistors are in on-state for $WL=0$

Write '1': BL=0V, WL=17V

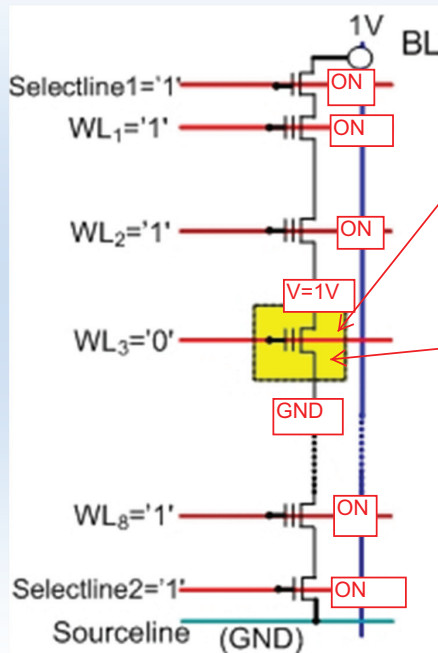
$\Rightarrow V_t$ no change $\approx -3V$
transistor not programmed

Write '0': BL=20V, WL=17V

$\Rightarrow V_t \approx +0.8V$
transistor programmed

NAND FLASH Memory

Read Operation



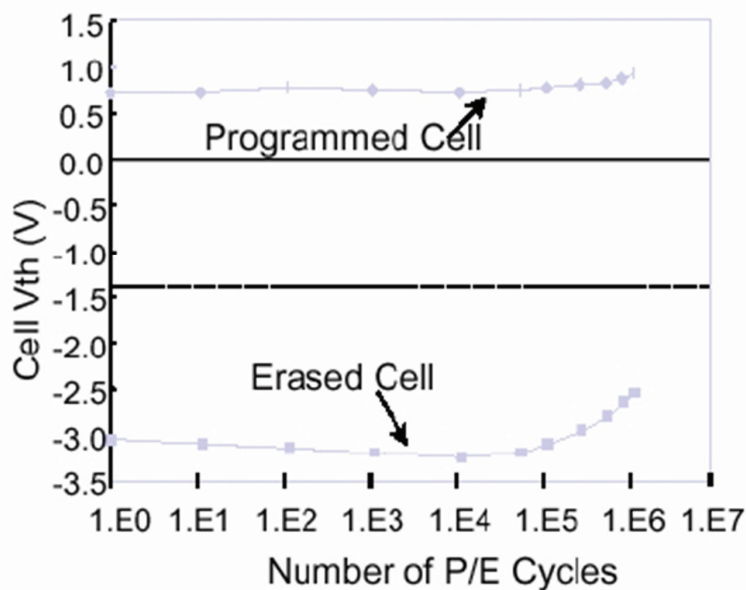
If selected floating-gate transistor (WL=0V) is not programmed, threshold voltage is negative and transistor is in on-state, resulting that the bitline BL is getting discharged! --> BL=LOW

If selected floating-gate transistor (WL=0V) is programmed, threshold voltage is positive and transistor is in off-state, resulting that the bitline BL remains on 1V! --> BL=HIGH

READ operation slow, because discharging of BL takes long time!

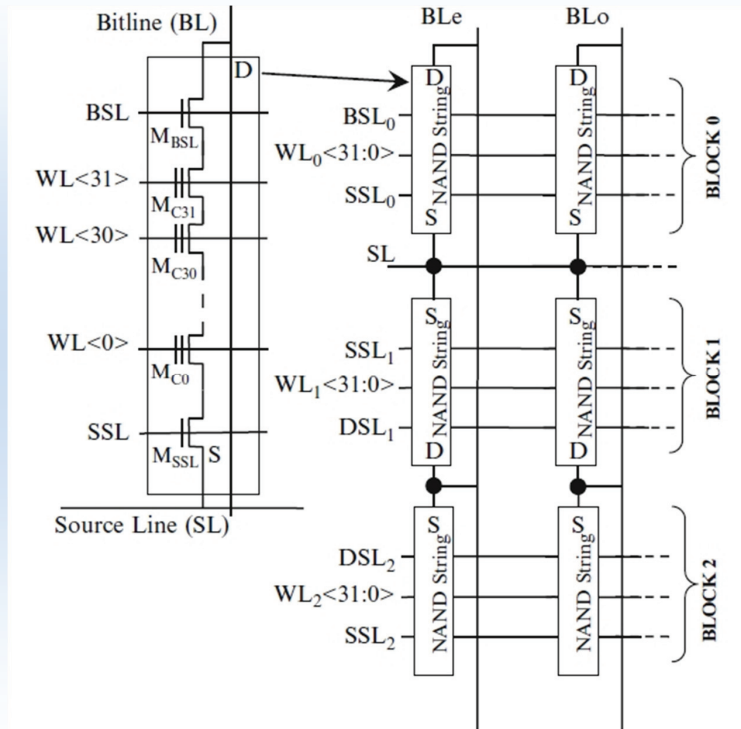
NAND FLASH Memory

Cell Vth Shift



[Samsung]

NAND FLASH Array



In WL direction, adjacent NAND strings share the same WL, SSL, BSL and SL

In BL direction, two consecutive strings share the bitline contact

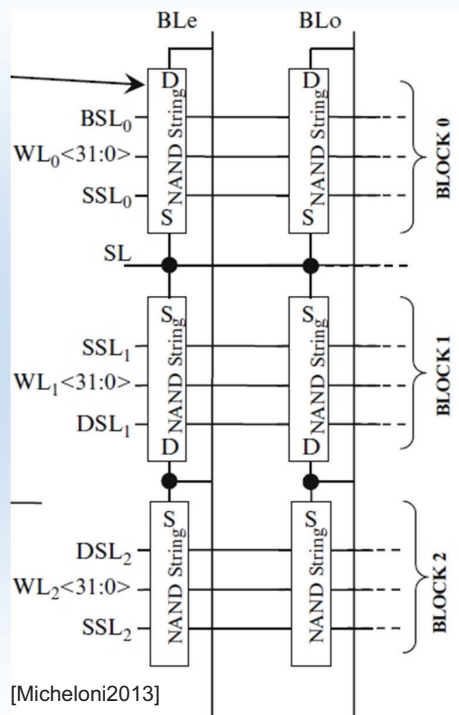
[Micheloni2013]

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NAND FLASH Array



[Micheloni2013]

All the NAND strings sharing the same group of WL's form a **block**

- BLOCK0 is made up by $WL_0<31:0>$;
- BLOCK1 is made up by $WL_1<31:0>$;
- BLOCK2 is made up by $WL_2<31:0>$.

A block is the smallest erasable unit!

Logical **pages** are made up of cells belonging to the same WL. A 4kB page means 32768 cells per WL.

A logical page is the smallest addressable unit for reading and writing!

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NOR vs. NAND-Flash

	SLC NAND-Flash	MLC NAND-Flash	MLC NOR-Flash
Capacity	512 Mbit – 128 Gbit	1 Gbit – 256 Gbit	128 Mbit – 2 Gbit
Datarate read	24 MB/s	18,6 MB/s	103 MB/s
Datarate write	8 MB/s	2,4 MB/s	0,47 MB/s
Erase	2 ms	2 ms	900 ms
Application	User Data Memory	User Data Memory	Program-Memory

Trends for FLASH Memory

Goal: Increase of density
Reduction of cost per bit

Single Level Cell (SLC): 1 bit per cell

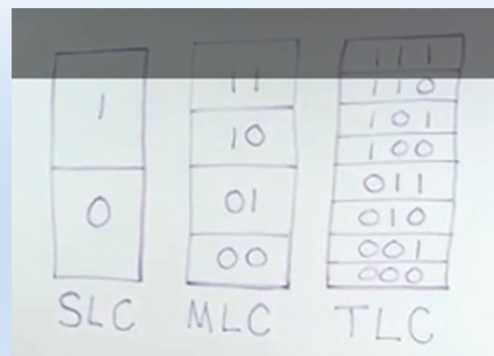
Multi Level Cell (MLC): 2 bit per cell

Triple Level Cell (TLC): 3 bit per cell

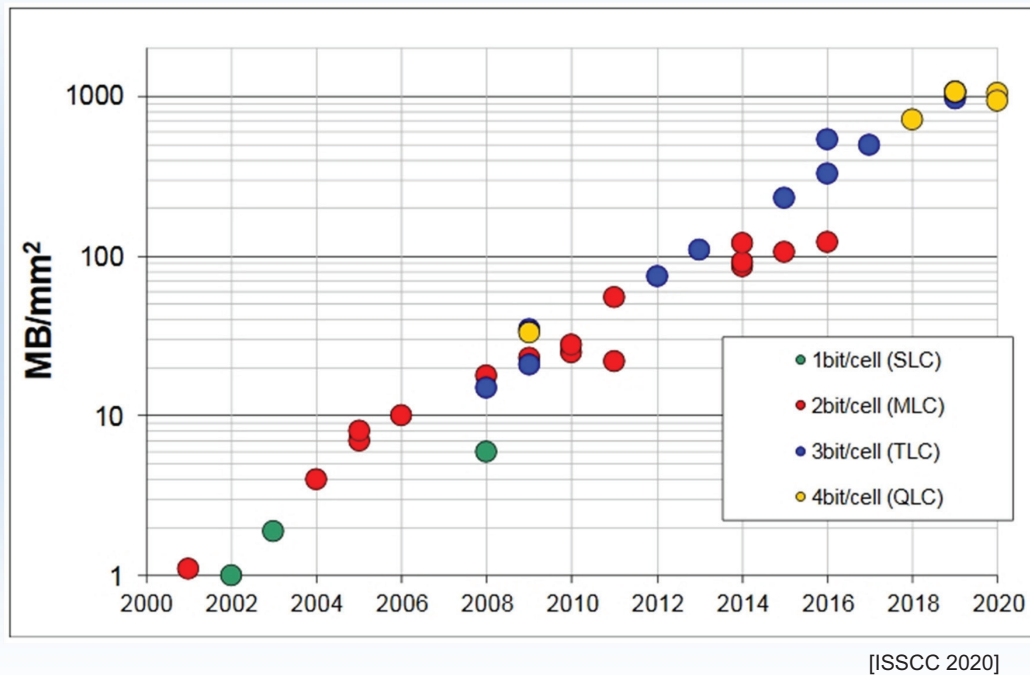
2017: 512Gbit TLC NAND-Flash (Samsung, Toshiba)

Quadrupel Level Cell (QLC): 4 bit per cell

2018: 1.33 Tbit QLC NAND-Flash, 96-Layers (Toshiba)



NAND FLASH Memory Trend



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SSD (NAND flash) replacing HDD

Parameter	Benchmark [A]	
	HDD [B]	NAND flash [B]
<i>Read/Write latency</i>	3-10 ms	~100μs (block erase ~1 ms)
Endurance (cycles)	unlimited	10 ³ -10 ⁵
<i>Retention</i>	>10 years	~10 years
ON power (W/GB)	0.003-0.05	~0.01-0.04
<i>Standby power</i>	~52%-69% of ON power [C]	<10% ON power
<i>Areal density</i>	~ 10 ¹¹ bit/cm ²	~ 10 ¹¹ bit/cm ²
Cost (\$/GB)	~0.1-1.0 [D]	2

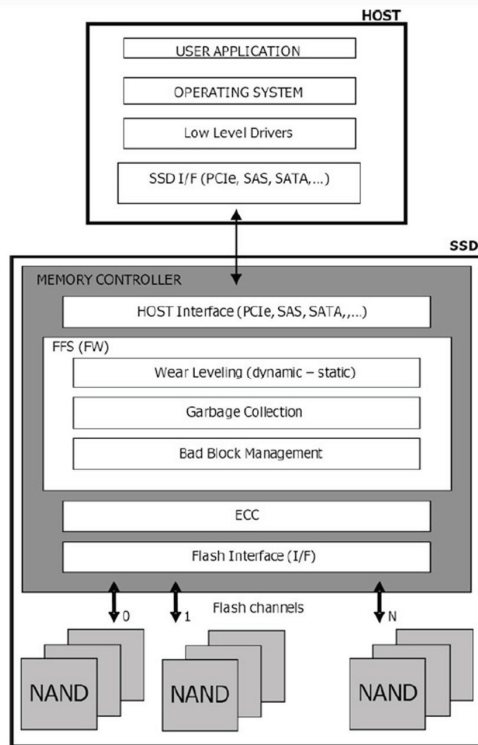
[ITRS-ERD 2015]

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SSD Memory Controller



[Micheloni2013]

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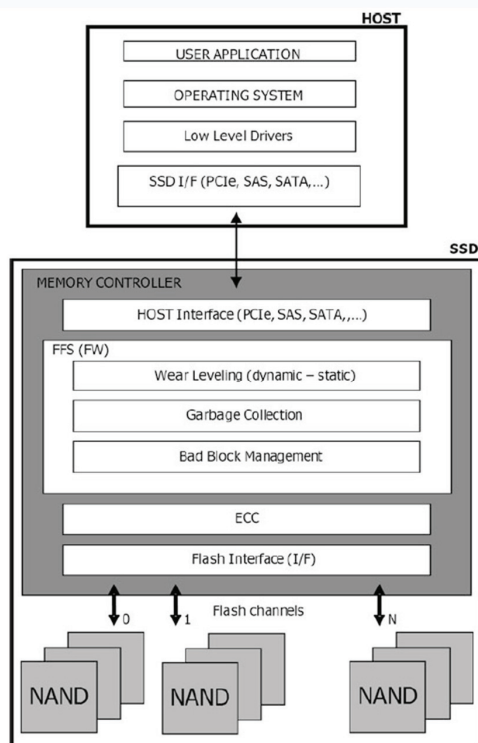
A memory controller has two fundamental tasks:

1. to provide the most suitable interface and protocol towards both the host and the Flash memories;
2. to efficiently handle data, maximizing transfer speed, data integrity and information retention.

Four parts:

- HOST Interface
- FFS (Flash File System)
- ECC (Error Correction Code)
- Flash Interface

SSD Memory Controller



[Micheloni2013]

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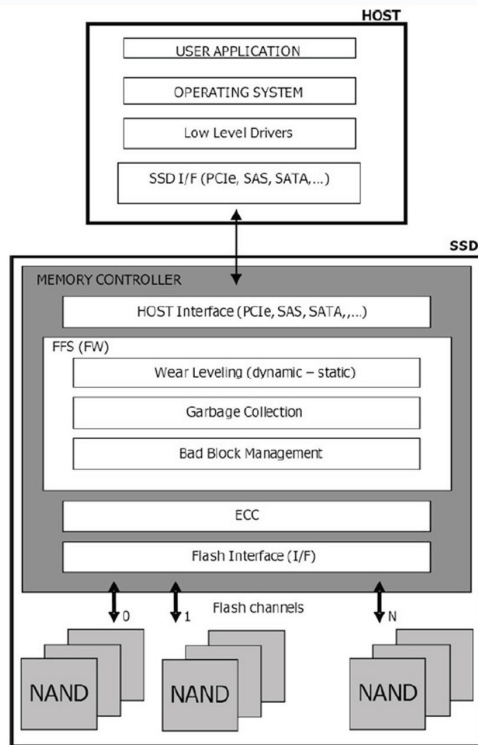
139

HOST Interface:
implements the required industry-standard protocol (PCIe, SAS, SATA, etc.) running firmware on μ C.

FFS:
the file system which enables the use of SSDs like magnetic disks (to build FAT). Main function are Wear Leveling, Garbage Collection and Bad Block Management.

ECC:
This task is typically executed by a specific hardware inside the memory controller.

SSD Memory Controller



Wear Leveling:
keep the number of write/erase cycles of each block as minimum and as uniform as possible! Strategie: Write to an erased block with the lowest erase count first.

Garbage Collection:
erases blocks with invalid information.

Bad Block Management:
creates a map of bad blocks during factory testing of the flash memory module. This map is updated during device life time whenever a block becomes bad!

[Michelsoni2013]

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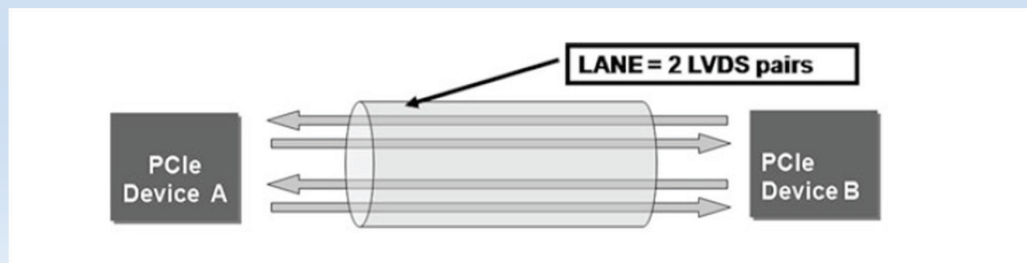
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SSD Interface Protocol

PCIe (Peripheral Component Interconnect Express) supports full-duplex communication between two endpoints.

Each pair of serial point-to-point connections, one for upstream, one for downstream, is called a **LANE**.



[Michelsoni2013]

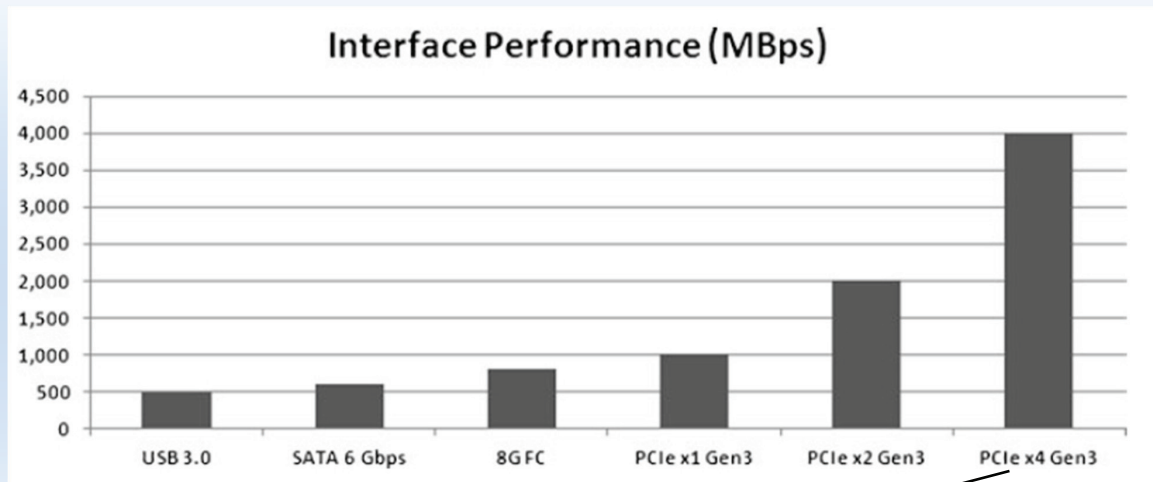
PCIe version	Year introduced	Throughput per lane
PCIe 1.0 (Gen1)	2003	250 MB/s
PCIe 2.0 (Gen2)	2007	500 MB/s
PCIe 3.0 (Gen3)	2010	1 GB/s
PCIe 4.0 (Gen4)	2017	2GB/s

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141

SSD Interface Performance



[Micheloni2013]

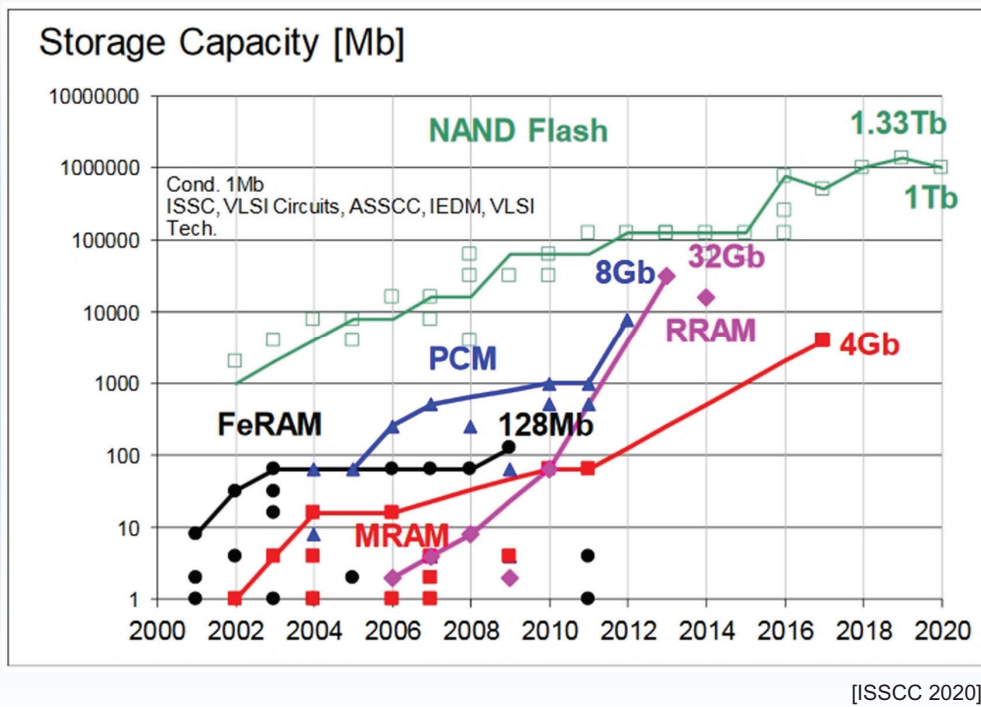
Max. 4 PCIe-LANES

Research: new nonvolatile memory devices

Challenges:

- fast read/write operations like SRAM
- nonvolatile with high density like FLASH
- infinite number of erasure operations, duration of storage > 20 years

Memory Capacity Trend of NV Memories



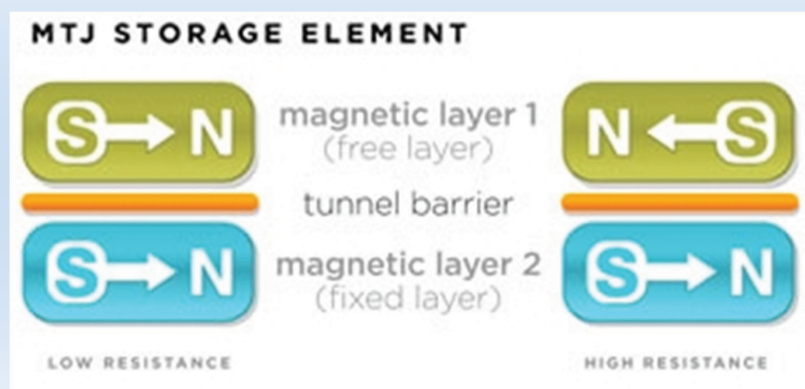
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MRAM

Idea: programmable resistance between two layers by changing the direction of a magnetic field in layer1



[Everspin Technologies Inc.]

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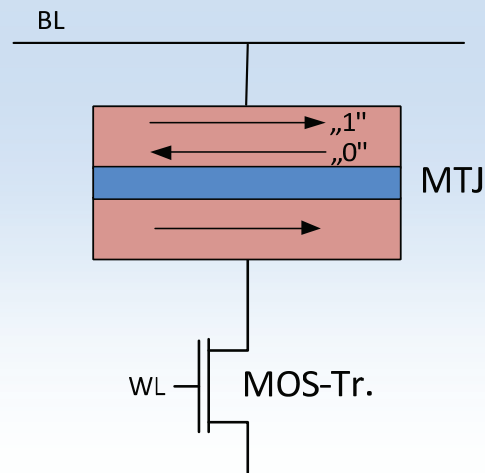
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145

MRAM

Concept:

Two plates are separated by an insulating layer. The bottom layer is pinned to a particular magnetic polarity. The other layer is programmable. When the two layers match, a "1" is stored. If they have opposite polarities, then a "0" is stored. Reading is accomplished by measuring the resistance of the memory cell.



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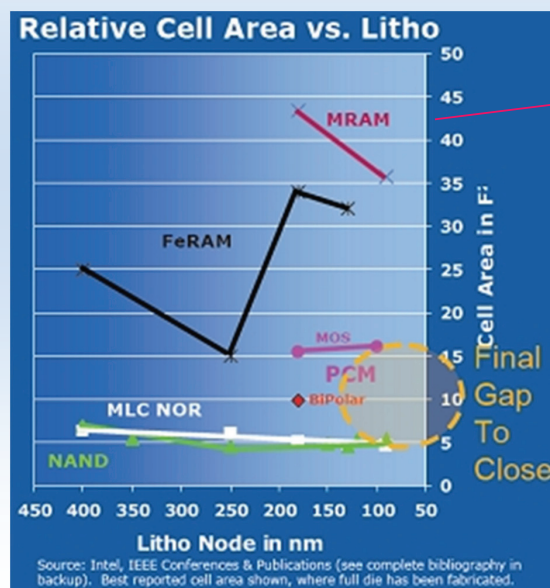
146

MRAM

Problem: large cell area, low memory capacity

2016: 256Mbit (Everspin Technologies Inc.)

2017: Prototype 4Gbit Spin-Transfer-Torque STT-MRAM (Toshiba)



cell area is 7-times larger than for FLASH

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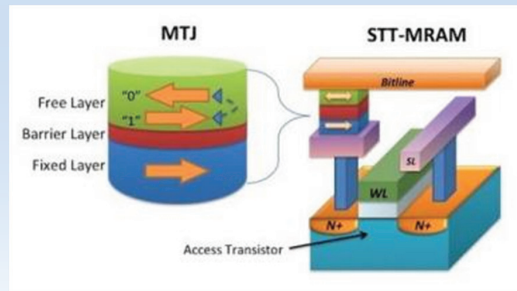
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147

STT-MRAM

Idea: Orientation of free magnetic layer can be changed using a spin-polarized current!

By passing a current through the fixed layer, one can produce a spin-polarized current. If this spin-polarized current is directed into a second, the free magnetic layer, angular momentum can be transferred to this layer, changing its magnetic orientation.



STT-MRAM

Advantages:

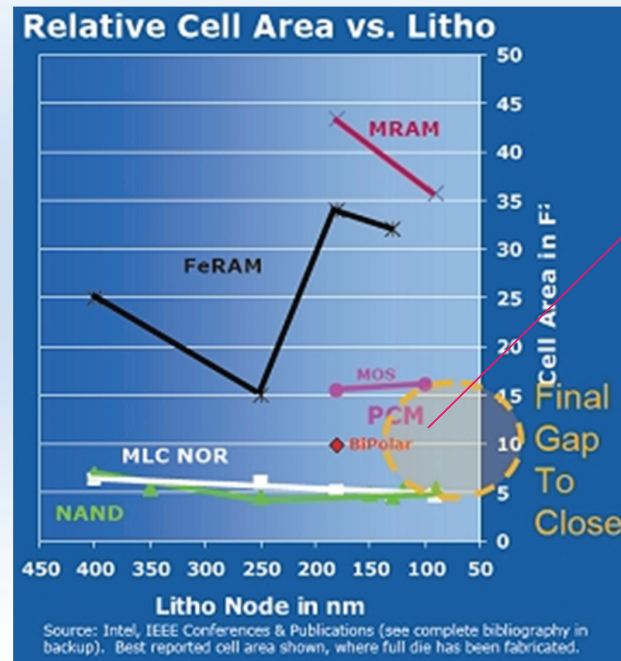
- 10^{12} write cycles
- fast read
- good scalability

Challenges:

- cost/bit is a major issue
- lack of MLC and 3D strategy
- high temp

[ITRS-ERD 2015]

Phase-Change Memory: PRAM, PCRAM, PCM



Challenge:
cell area is still 3-
times of FLASH!

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Phase-Change Memory: PRAM, PCRAM, PCM

Phase-Change Memory uses a material which could change from amorphous structure to polycrystalline structure and vice versa

Non-volatile memory

Structure	Sample	Properties
Amorphous		• Short-range atomic Order • High reflectivity • High resistivity
Polycrystalline		• Long-range atomic Order • Low reflectivity • Low resistivity

[Source: Intel, Ovonyx]

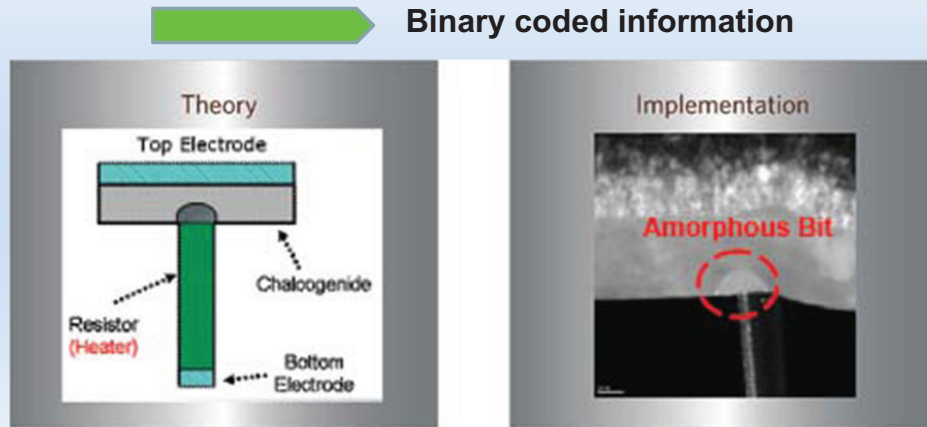
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Phase-Change Memory: PRAM, PCRAM, PCM

Phase-Change Memory cell shows a high resistance when in amorphous state and a low resistance when in polycrystalline state



[Source: Numonyx (a Micron Technology Inc. company)]

2012: Samsung presents 20nm 1.8V 8Gb PRAM

PRAM

Advantages:

- 10^8 write cycles (endurance x100 vs. Flash)
- MLC and 3D strategy possible
- very good scalability

Challenges:

- power consumption
- speed

[ITRS-ERD 2015]

Resistive RAM: ReRAM, RRAM

It works by changing the resistance across a dielectric solid-state material

2013: 32-Gb ReRAM Memory Device in 24-nm technology

Problems: high write latency (slow) and high write power consumption

but: **R/W faster than PRAM!**

	SRAM	EDRAM	STT-RAM	RRAM	PCM
Cell size (F^2)	120 – 200	60 – 100	6 – 50	4 – 10	4 – 12
Write Endurance	10^{16}	10^{16}	4×10^{12}	10^{11}	$10^8 - 10^9$
Speed (R/W)	Very fast	Fast	Fast/slow	Fast/slow	Slow/very slow
Leakage Power	High	Medium	Low	Low	Low
Dynamic Energy (R/W)	Low	Medium	Low/high	Low/high	Medium/high
Retention Period	N/A	30 – 100 μ s	N/A (unless relaxed)	N/A	N/A

[Mittal15]

RRAM

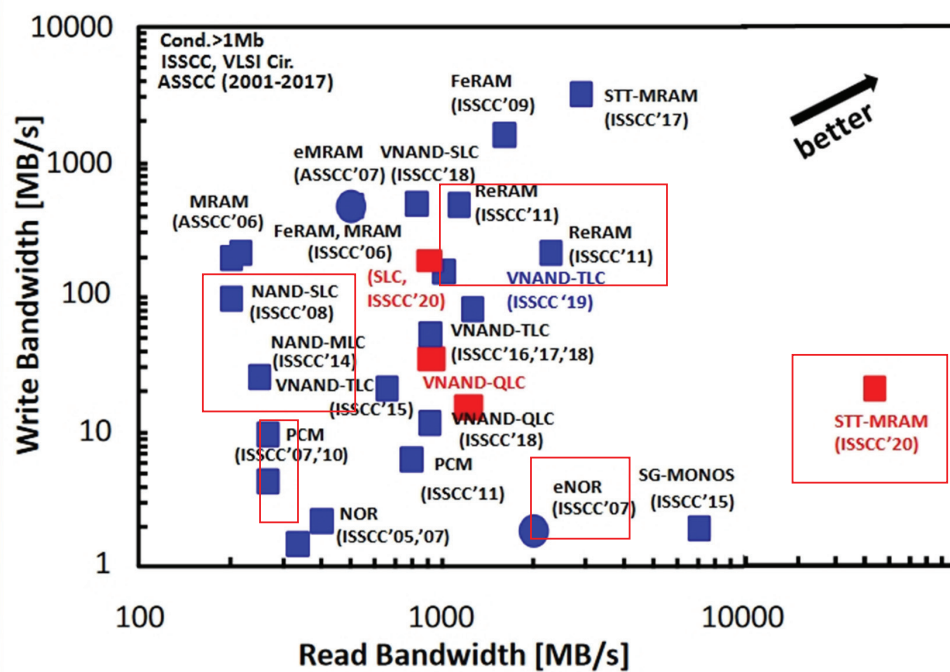
Advantages:

- 10^{11} write cycles (very high endurance)
- fast read
- very good scalability

Challenges:

- power consumption
- write speed

Read/Write Bandwidth of NV Memories



[ISSCC 2020]

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156

Summary of new NV Memories

	Prototypical (Table ERD3)		
Parameter	FeRAM	STT-MRAM	PCRAM
Scalability	☹	☹	😊
MLC	☹	☹	😊
3D integration	☹	☹	😊
Fabrication cost	☹	☹	☹
Retention	☹	☹	☹
Latency	😊	😊	☹
Power	😊	☹	☹
Endurance	😊	😊	☹
Variability	😊	☹	☹

[ITRS-ERD 2015]

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157

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